

Stakeholder

Question

Catalogue

STAKEHOLDER 21

Access-control administrators

Evidence-bounded decision support for request validation, credential lifecycle, configuration, fault diagnosis, least-privilege assurance, incident response and privacy-preserving reporting.

OPERATIONAL DECISIONS, BOUNDED EVIDENCE

A standalone catalogue for validating authority, assembling least-privilege access, tracing credential and configuration dependencies, diagnosing technical faults and reporting service performance without turning access-control records into movement surveillance.

80 STAKEHOLDER QUESTIONS **8** OPERATIONAL CATEGORIES **6** FIELDS IN EVERY RECORD

CATALOGUE MAP

1 Identity and role-based access requests AC-001 to AC-010	2 Credential issue, replacement and lifecycle AC-011 to AC-020
3 Doors, zones, permission groups and dependencies AC-021 to AC-030	4 Schedules, temporary access and visitors AC-031 to AC-040
5 Reader, controller and integration faults AC-041 to AC-050	6 Reviews, audit trails and least privilege AC-051 to AC-060
7 Incidents, suspension, revocation and recovery AC-061 to AC-070	8 Privacy, retention, governance and service reporting AC-071 to AC-080

Evidence posture: Every sensor, integration and analytical capability is proposed until current deployment, mapping, commissioning, permission and health are independently verified. The May 2021 floor plan is a historical spatial reference only and cannot establish current room use, access, services, hazards, routes or closures.

Scope and operating stance

What this catalogue supports - and where it must stop

THE CATALOGUE SUPPORTS

- Validate whether a request has minimum current evidence and accountable approvals.
- Trace identity, role, credential, group, schedule, opening, controller and integration dependencies.
- Diagnose bounded technical faults using controlled tests, measured coverage and explicit uncertainty.
- Prepare least-privilege reviews, incident status and privacy-preserving service reports.
- Expose stale, conflicting, unmapped or unverified evidence before any authorised action.

THE SERVICE NEVER DOES

- Grant, modify, suspend, revoke, restore or expire access, or control a door, lift or emergency mode.
- Infer identity, intent, attendance, behaviour, route, occupancy or physical passage from transactions.
- Disclose credential or movement records without a verified purpose, permission and authorised case role.
- Handle passwords, PINs, private keys, cryptographic material or unrestricted configuration exports.
- Override identity, HR, Student Records, Security, sponsor, incident-command or Data Protection decisions.

AUTHORITY TAKES PRECEDENCE

- HR, Student Records and Identity and Access Management define identity, affiliation and identity linkage.
- Sponsors, managers, training owners and area owners define duties, prerequisites and accountable purpose.
- Security owns approved permission groups, exception routes, operating modes and access policy.
- Competent fire, lift, electrical, accessibility and safety professionals own safety-critical decisions.
- Data Protection, Legal, Records Management and Information Security own lawful use, disclosure and retention.

Evidence posture: proposed capability, verified reality

- 1 A proposed source becomes usable only after asset or data mapping, commissioning, purpose permission, functional validation, clock alignment and current health are verified.
- 2 A credential transaction records a system decision at a reader. It is not proof of identity, passage, attendance, safety, security effectiveness or intent.
- 3 A missing event is not evidence that a person did not attend, that an opening did not operate or that a controller enforced the intended state.
- 4 Every answer carries source owner, effective, observed or approved, and retrieved times; coverage; aggregation; uptime; gaps; drift; provenance; staleness; conflict and uncertainty.
- 5 When current evidence is incomplete or conflicts, the safe default is to keep access unchanged and route the exact issue to the named authoritative owner.

Field guide

Six substantive fields in every question record

1

Stakeholder question

A natural access-administration decision identifying the case, object, interval or operational choice, rather than requesting generic information.

2

Sensors and telemetry

Proposed modality and location; asset and temporal coverage; sampling or aggregation; calibration or controlled validation; clock alignment; uptime, gaps, drift, provenance and limits.

3

Authoritative supporting data

Named source owners and precedence; effective, observed or approved, and retrieved times; permissions; provenance; staleness rules; and explicit unresolved conflicts.

4

Analysis required

An explicit lookup, observation, calculation, estimate, forecast, comparison, diagnosis or recommendation with decision interval, minimum evidence, method, uncertainty and safe failure.

5

Answer boundary

What the service must not infer, disclose, promise, certify, authorise, execute or control; and which owner or competent professional must decide.

6

Why this is useful

The concrete administrative decision, escalation, test, evidence pack, queue action or owner consultation that the answer enables.

SAFE DEFAULT

When minimum evidence, permission, mapping or current health is absent, the answer is UNKNOWN or CONFLICTED. Access remains unchanged and the exact gap is routed to its authoritative owner.

Evidence, time and permission grammar

How every answer is bounded

EVIDENCE CHAIN

1

Authority

Current owner decisions, policy, affiliation, sponsorship, prerequisites and approved procedure.

2

Configuration

Versioned identity, credential, group, schedule, opening and integration mappings.

3

Observation

Health, transaction and controlled-test evidence with measured coverage and quality limits.

4

Analysis

One bounded operation reporting state, uncertainty and a safe non-executing next step.

THREE TIMES

Effective

When a status, approval, schedule or policy is intended to apply.

Observed / approved

When telemetry, a transaction, a controlled test or a human decision was recorded.

Retrieved

When the service obtained the record. Retrieval time never replaces the other two.

ANSWER STATES

VERIFIED

Minimum evidence is current, consistent and within measured coverage.

PARTIALLY VERIFIED

Some evidence is sound, but stated coverage or quality gaps remain.

CONFLICTED

Current sources disagree and their owners must resolve the difference.

UNKNOWN

Minimum evidence, permission, mapping, health or provenance is absent.

OPERATION TYPES

Lookup / observation

Retrieve authority or describe current measured state.

Calculation / estimate

Derive exact values where possible; bound any inferred quantity.

Forecast / comparison

Project from declared assumptions or compare like versions and periods.

Diagnosis / recommendation

Rank supported causes or next steps; never execute or authorise.

PERMISSION POSTURE

Use person-level records only for a verified purpose and authorised case role. Prefer configuration, case status and safe aggregates. Keep conflicts visible. When the minimum evidence is absent, leave access unchanged and route the exact gap to its owner.

Identity and role-based access requests

AC-001 to AC-010

AC-001

QUESTION 1 OF 10

STAKEHOLDER QUESTION

Which minimum approved permission package should be prepared for this verified starter on the authorised effective date?

SENSORS AND TELEMETRY

Proposed evidence and location: request-workflow events, identity-directory synchronisation status, role-template versions and dry-run entitlement results at the case, group and affected controller; no environmental sensor is required. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the verified identity record, requested role and location, direct and inherited permission groups, prerequisites and every mapped controller expected to receive them.

Timing/aggregation: at request creation, after each material source change and immediately before hand-off to an authorised implementer; connector health at its configured heartbeat, normally minutes rather than hours; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate source-to-field mapping with controlled test identities, compare configuration hashes and confirm controller acknowledgements without using another person as a template; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: future-dated appointments, unmapped openings, manual exceptions, offline controllers and directory lag can leave the package incomplete even when the workflow appears complete.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: HR or Student Records own affiliation; Identity and Access Management owns identity linkage; the manager or academic sponsor owns duties; training and area owners own prerequisites; Security owns access policy and role matrices. A verified active or future-effective affiliation, accountable approval and the current Security-approved role matrix outrank email wording, copied entitlements, historic practice and the May 2021 floor plan. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: An approval or case record is stale when expired, superseded, outside its effective window, or missing a current owner/version; a connector state beyond the owner-approved freshness limit is UNKNOWN. Show conflicting values side by side and route them to the named identity, sponsor, prerequisite or Security owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: comparison plus bounded recommendation. Decision interval: on submission, after a material change and immediately before implementation. Minimum evidence: stable identity ID, effective start date, approved role and location, accountable sponsor, current prerequisite status, current template version and mapped target controllers. Method: intersect approved role groups with prerequisites, exclusions, dates and route dependencies; list only the smallest supported set and every unresolved gap. Uncertainty: ambiguous role names, overlapping appointments, inherited groups, future-effective records, connector delay and unmapped dependencies. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: leave access unchanged, return UNKNOWN or CONFLICTED and route the exact missing evidence to its owner.

ANSWER BOUNDARY

It must not create or choose an identity, infer duties from a title alone, copy a colleague's permissions or treat a planned start as confirmed attendance. It must not decide identity, affiliation, role legitimacy, sponsorship or prerequisite outcomes; expose protected person or credential records beyond the authorised case; authorise or execute access changes; promise entry; or treat a transaction as proof of passage.

WHY THIS IS USEFUL

The administrator can prepare the smallest defensible starter package and return incomplete cases before credential issue.

AC-002

QUESTION 2 OF 10

STAKEHOLDER QUESTION

For this role or department transfer, which entitlements are justified to add, retain, remove-candidate or refer to an owner?

SENSORS AND TELEMETRY

Proposed evidence and location: before-and-after entitlement snapshots, role-change workflow events, group graph versions and controller distribution acknowledgements at the identity, permission-group and opening level. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: all direct, nested and temporary entitlements linked to the identity, including doors, lift floors, schedules and exceptions affected by the old or new role. Timing/aggregation: at the submission, at the authoritative effective time, after each approved change and after controller propagation; show deltas by case and object, not movement history; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: verify snapshot completeness against the configuration inventory, match group versions by hash and test the delta logic with synthetic cases that include nested groups and overlapping roles; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: off-workflow changes, unresolved concurrent appointments, local controller caches and unrecorded exceptions can make a clean-looking delta misleading.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: HR or Student Records own the effective role change; old and new managers own duty evidence; area and training owners own prerequisites; Security owns permission groups and exception policy. The effective role-change record and current approved access matrix outrank the previous entitlement state, informal retention requests and observed use. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: An approval or case record is stale when expired, superseded, outside its effective window, or missing a current owner/version; a connector state beyond the owner-approved freshness limit is UNKNOWN. Show conflicting values side by side and route them to the old and new authoritative role owners and Security; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: comparison and recommendation. Decision interval: at submission, immediately before change and after propagation evidence is available. Minimum evidence: verified old and new roles, effective time, current entitlement snapshot, current group graph, area approvals, prerequisites and exception end dates. Method: calculate direct and inherited entitlement deltas, separating add, retain, remove-candidate and owner-review items without executing any action. Uncertainty: overlapping appointments, inherited permissions, stale snapshots, incomplete propagation and duties that legitimately continue across the change. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: keep the current state, mark each unsupported delta UNKNOWN or CONFLICTED and route it for accountable decision.

ANSWER BOUNDARY

It must not decide which role is legitimate, remove access because it appears unused, or assume that a title change cancels continuing duties. It must not decide identity, affiliation, role legitimacy, sponsorship or prerequisite outcomes; expose protected person or credential records beyond the authorised case; authorise or execute access changes; promise entry; or treat a transaction as proof of passage.

WHY THIS IS USEFUL

The administrator can prepare an accountable bidirectional change list that prevents privilege accumulation and accidental service loss.

Identity and role-based access requests

AC-001 to AC-010

AC-003

QUESTION 3 OF 10

STAKEHOLDER QUESTION

How should concurrent staff, student and sponsored-project affiliations be combined without widening access beyond each approved duty?

SENSORS AND TELEMETRY

Proposed evidence and location: linked identity and affiliation records, role-template assignments, conflict-rule evaluations and credential-to-identity binding status in the identity and access platforms; no physical sensor is required. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: every active and future-effective affiliation for the same verified identity, its sponsors, locations, dates, approved role packages and conflicts between them.

Timing/aggregation: whenever an affiliation starts, ends or changes and immediately before an authorised access decision; summarise by affiliation and entitlement source; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate identity-key linkage with approved test cases, detect duplicate person records and verify conflict-rule versions against the policy owner; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: similar names, delayed end dates, duplicate identities, sponsor changes and broad template unions can create false completeness or excess privilege.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Identity and Access Management owns person-record linkage; HR and Student Records own their respective affiliations; project sponsors own sponsored purpose; Security owns combination and conflict rules. Verified source-system identifiers and effective affiliation dates outrank name matching, email aliases, local spreadsheets and assumptions that all roles may simply be combined. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A feed is stale after its owner-approved freshness limit; if no limit is documented, more than two missed scheduled updates, a schema change, or an unexplained count discontinuity makes the affected result UNKNOWN. Show conflicting values side by side and route them to Identity and Access Management, the affected affiliation owners and Security; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, comparison and bounded recommendation. Decision interval: on each affiliation event and before implementation. Minimum evidence: one stable identity link, current and future affiliations with dates, accountable sponsors, applicable role templates, conflict rules and existing entitlements. Method: attribute each entitlement to a specific approved affiliation, calculate the least-privilege union, apply documented exclusions and expose unresolved overlaps. Uncertainty: identity-link ambiguity, future-effective roles, conflicting end dates, inherited groups and rules that do not cover the observed combination. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: do not combine packages; leave access unchanged and require the authoritative owners to resolve the identity or role conflict.

ANSWER BOUNDARY

It must not decide that two records are the same person from names alone, choose the dominant affiliation, or treat concurrent roles as permission to aggregate every entitlement. It must not decide identity, affiliation, role legitimacy, sponsorship or prerequisite outcomes; expose protected person or credential records beyond the authorised case; authorise or execute access changes; promise entry; or treat a transaction as proof of passage.

WHY THIS IS USEFUL

The administrator can build a traceable multi-role package without granting the broad union of every possible role.

AC-004

QUESTION 4 OF 10

STAKEHOLDER QUESTION

Which approved role template, or smallest justified combination, covers duties across these teams and buildings?

SENSORS AND TELEMETRY

Proposed evidence and location: versioned role-template and permission-group graphs, door and lift dependency mappings, assignment counts and dry-run change results in the PACS configuration repository; no live occupancy sensing is required. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the named duties, teams and sites in the approved request, all direct and inherited groups in candidate templates, and mapped route dependencies to the requested destinations. Timing/aggregation: at request review and after any template, door-map or dependency change; aggregate by candidate template, site and excess permission set; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate graph edges against commissioned asset identifiers, test set calculations with known template examples and verify configuration hashes and clocked snapshot times; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: template names can conceal different versions, broad campus groups, unmapped assets, temporary exceptions and route dependencies not represented in a historical plan.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: managers or sponsors own the stated duties; building and area owners own destination approval; Security owns templates and group composition; Estates and lift owners own current asset mappings. Approved duties and current Security-owned template versions outrank job-title inference, prior holders, local custom and the May 2021 floor plan. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to the duty, area, asset-mapping and Security owners; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: comparison, calculation and recommendation. Decision interval: during request design and immediately before an authorised implementation decision. Minimum evidence: approved duty list, site and destination scope, current template graph, group-to-asset mapping, route dependencies and exclusion rules. Method: solve a constrained minimum-cover comparison: cover every approved destination and prerequisite while identifying permissions added beyond the stated duties. Uncertainty: unmapped openings, incomplete route data, shared groups, nested permissions and duties that are not represented by a current template. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return no template recommendation where coverage or excess privilege cannot be measured; require owner-designed access rather than broad fallback groups.

ANSWER BOUNDARY

It must not infer duties from a department name, decide that broad building access is operationally necessary, or create a new role template. It must not decide identity, affiliation, role legitimacy, sponsorship or prerequisite outcomes; expose protected person or credential records beyond the authorised case; authorise or execute access changes; promise entry; or treat a transaction as proof of passage.

WHY THIS IS USEFUL

The administrator can select or propose the narrowest defensible template combination instead of granting convenient broad access.

Identity and role-based access requests

AC-001 to AC-010

AC-005

QUESTION 5 OF 10

STAKEHOLDER QUESTION

Is this access request ready for administration, and which accountable owner must supply each missing item?

SENSORS AND TELEMETRY

Proposed evidence and location: structured workflow state, electronic approval status, identity and affiliation sync health, prerequisite feed status and attachment provenance at the individual case; no door sensor is required. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the complete request chain from requester through sponsor, area, training and Security decisions, including every conditional prerequisite referenced by the requested group. Timing/aggregation: on submission, after each approval or source update and immediately before case hand-off; show a checklist and missing-evidence count rather than exposing unrelated records; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate workflow transitions with approved test cases, verify signer-role mapping and detect missing, duplicate, expired or superseded approvals; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: an apparently complete form can still rely on stale identity data, an invalid delegation, an unmapped group or an approval that does not cover the requested scope or dates.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the requester owns factual request details; Identity, HR or Student Records own identity and affiliation; sponsors, area and training owners own their approvals; Security owns final access-policy sufficiency. Current authoritative source records and scope-matched approvals outrank free-text statements, forwarded email, inherited workflow status and prior similar cases. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: An approval or case record is stale when expired, superseded, outside its effective window, or missing a current owner/version; a connector state beyond the owner-approved freshness limit is UNKNOWN. Show conflicting values side by side and route them to the owner of each missing or conflicting checklist item; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup and rules-based validation. Decision interval: continuously while the case is open and immediately before authorised action. Minimum evidence: stable case and identity IDs, requested doors or groups, effective dates, purpose, accountable sponsor, required approval chain and current prerequisite evidence. Method: evaluate each mandatory rule against current evidence, distinguish missing from conflicting items and identify the accountable owner for every failure. Uncertainty: conditional rules, changed scope, invalid delegations, ambiguous attachments, integration delay and policy rules without machine-readable form. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: mark the case NOT READY, make no change and return only the precise missing or conflicting requirements.

ANSWER BOUNDARY

It must not treat workflow completion as substantive approval, invent missing consent, interpret an attachment beyond its authorised metadata, or bypass an accountable owner. It must not decide identity, affiliation, role legitimacy, sponsorship or prerequisite outcomes; expose protected person or credential records beyond the authorised case; authorise or execute access changes; promise entry; or treat a transaction as proof of passage.

WHY THIS IS USEFUL

The administrator can reject incomplete cases consistently and avoid spending time diagnosing generic workflow failures.

AC-006

QUESTION 6 OF 10

STAKEHOLDER QUESTION

HR or Student Records, the identity platform and the sponsor disagree on status or end date - which conflict blocks processing?

SENSORS AND TELEMETRY

Proposed evidence and location: source-record snapshots, connector lineage, field-level discrepancy flags and synchronisation job health across HR, Student, IAM and sponsor workflow systems; no physical sensing is required. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the specific identity, affiliation, sponsor and requested access case, including all source values, effective dates, record versions and the transformations that produced the PACS view. Timing/aggregation: when a discrepancy is first detected, after any source correction and immediately before processing; present source values side by side, never as a blended value; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate connector mappings against controlled reference records, compare source and target checksums and verify clock offset and queue delay for each hop; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: source systems may use different business meanings, planned dates, grace periods or update cycles; technical freshness does not establish which business value is correct.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: HR owns staff affiliation, Student Records owns student status, IAM owns identity linkage, the sponsor owns sponsored purpose and Security owns the access decision based on resolved evidence. The designated authoritative owner for each field outranks the directory copy, sponsor recollection, PACS cache, email and historical spatial records. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A feed is stale after its owner-approved freshness limit; if no limit is documented, more than two missed scheduled updates, a schema change, or an unexplained count discontinuity makes the affected result UNKNOWN. Show conflicting values side by side and route them to the authoritative owner for the disputed field, with Identity and Security coordination; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: comparison and conflict diagnosis. Decision interval: at detection, after correction and before any authorised change. Minimum evidence: stable identity link, each conflicting source record with effective and retrieval times, connector lineage, owner designation and the request scope. Method: classify whether the difference is expected latency, semantic mismatch, future-effective data or a true owner conflict; do not choose a value. Uncertainty: clock skew, queued updates, different definitions of end date, duplicate identities, manual overrides and incomplete source-owner documentation. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return CONFLICTED, freeze the case and route a field-specific evidence bundle to the authoritative owner.

ANSWER BOUNDARY

It must not select the most recent-looking value, decide employment or student status, or overwrite an authoritative source from PACS data. It must not decide identity, affiliation, role legitimacy, sponsorship or prerequisite outcomes; expose protected person or credential records beyond the authorised case; authorise or execute access changes; promise entry; or treat a transaction as proof of passage.

WHY THIS IS USEFUL

The administrator can stop unsafe processing early and send the right discrepancy to the right owner with reproducible evidence.

Identity and role-based access requests

AC-001 to AC-010

AC-007

QUESTION 7 OF 10

STAKEHOLDER QUESTION

Are all induction, training, competency and area-approval prerequisites effective for the requested restricted-zone interval?

SENSORS AND TELEMETRY

Proposed evidence and location: prerequisite status feeds from approved training, competency, permit and area-approval systems, plus permission-group dependency metadata; no environmental or behavioural sensing is required. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the requested zone and every prerequisite explicitly linked to its current access rule, for the named identity and proposed effective period. Timing/aggregation: at request review, after a prerequisite change and immediately before implementation; display current/expired/missing status and dates, not course content or assessment detail; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate identifier mapping between training and access systems with known test records, verify rule versions and confirm that completion and expiry dates use aligned clocks and time zones; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a training feed may be delayed, a course title may not equal the required competency, and completion data cannot establish present practical competence or safe working conditions.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: training providers own completion records; competent area, laboratory and safety owners own required competency and approval; Security owns the access rule; HR or Student Records own affiliation. The current area-owner prerequisite matrix and authoritative completion records outrank self-declaration, certificates without provenance, role assumptions and old room labels. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A feed is stale after its owner-approved freshness limit; if no limit is documented, more than two missed scheduled updates, a schema change, or an unexplained count discontinuity makes the affected result UNKNOWN. Show conflicting values side by side and route them to the training, competency, area or safety owner responsible for the failed prerequisite; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup and comparison. Decision interval: on request, on prerequisite change and before authorised implementation. Minimum evidence: verified identity, requested zone and dates, current prerequisite rule version, authoritative completion or approval records and any documented exception authority. Method: match each required prerequisite to a current evidence record, calculate validity across the requested period and expose unmatched or expiring items. Uncertainty: synonyms between course and competency names, delayed feeds, future expiry during the access period, unrecorded exemptions and unmapped zone rules. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return UNKNOWN or NOT READY, leave access unchanged and require the competent owner to confirm the missing prerequisite.

ANSWER BOUNDARY

It must not certify competence, infer safe practice from course completion, waive a prerequisite or decide whether an area is currently safe or operational. It must not decide identity, affiliation, role legitimacy, sponsorship or prerequisite outcomes; expose protected person or credential records beyond the authorised case; authorise or execute access changes; promise entry; or treat a transaction as proof of passage.

WHY THIS IS USEFUL

The administrator can hold or route restricted-area requests before a credential is configured against incomplete prerequisites.

AC-008

QUESTION 8 OF 10

STAKEHOLDER QUESTION

Does this scope genuinely require a formal exception, or can it be met by a narrower approved template?

SENSORS AND TELEMETRY

Proposed evidence and location: structured requested-scope data, current role-template and group graphs, template-match results, exception-workflow state and hard-expiry fields in the access administration systems. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the requested identity or cohort, destinations, schedules, dates and every candidate standard template, including permissions that each candidate would add beyond the request. Timing/aggregation: at request design, after any scope or template change and before approval routing; aggregate match, gap and excess counts by template; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate template graph versions and asset mappings, test the match algorithm with known standard and exception cases and verify that expiry fields propagate to every nested path; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: an unmatched request may reflect an unmapped asset or obsolete template rather than a valid exception; a close match may still grant materially broader access.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the sponsor owns the business purpose; area owners own destinations; Security owns standard templates, exception criteria and approval route; Records Management owns required exception evidence. Current Security policy and owner-approved scope outrank precedent, convenience, another person's entitlements and historical room use. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to Security and the relevant sponsor and area owners; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: comparison and recommendation. Decision interval: on submission and whenever the requested scope or governing template changes. Minimum evidence: approved purpose, exact destinations and dates, current template graph, mapped assets, exception criteria, accountable owner and proposed expiry. Method: compare the request with every current standard template, quantify uncovered need and excess privilege, and identify the documented exception route without approving it. Uncertainty: incomplete asset maps, shared groups, policy gaps, ambiguous purpose and nested permissions whose expiry cannot be proved. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: do not fall back to the nearest broad template; return NOT READY and require a formal, owner-approved exception design.

ANSWER BOUNDARY

It must not declare an exception justified, create or approve the exception, or use historical precedent as authority. It must not decide identity, affiliation, role legitimacy, sponsorship or prerequisite outcomes; expose protected person or credential records beyond the authorised case; authorise or execute access changes; promise entry; or treat a transaction as proof of passage.

WHY THIS IS USEFUL

The administrator can distinguish genuine exception work from poor mapping and prevent broad-template shortcuts.

Identity and role-based access requests

AC-001 to AC-010

AC-009

QUESTION 9 OF 10

STAKEHOLDER QUESTION

Is this external collaborator case bounded by a verified identity, accountable sponsor, approved purpose, exact zones and hard end date?

SENSORS AND TELEMETRY

Proposed evidence and location: identity-proofing outcome status, collaborator or visitor registry records, sponsor workflow, purpose and date fields, credential lifecycle state and integration health; no identity document contents are ingested. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the external identity case, sponsor, organisational affiliation, requested zones, prerequisites, credential and every effective start and end date. Timing/aggregation: at application, after sponsor or affiliation change, at credential issue and before any extension; show status by case and requirement; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate source identifiers and sponsor-role mapping with controlled cases, verify date propagation into groups and controller caches, and check for duplicate active collaborator records; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a proofing-status flag does not reveal how identity was established, sponsor approval does not establish legal eligibility, and end-date mismatches can create unintended continuation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the approved identity-proofing service owns verification outcome; the external-collaborator registry owns affiliation; the sponsor owns purpose; area and training owners own prerequisites; Security owns access policy. Authoritative verification outcome, current accountable sponsorship and explicit effective dates outrank email introductions, local lists, prior visits and name similarity. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: An approval or case record is stale when expired, superseded, outside its effective window, or missing a current owner/version; a connector state beyond the owner-approved freshness limit is UNKNOWN. Show conflicting values side by side and route them to the identity-proofing, sponsor, affiliation, area or Security owner responsible for the unresolved item; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup and rules-based validation. Decision interval: at intake, before issue, on material change and before extension. Minimum evidence: stable external identity ID, verification outcome, accountable sponsor, approved purpose, exact scope, prerequisites and non-null start and end dates. Method: validate each requirement, check date consistency and duplicate bindings, and calculate whether the requested access remains entirely within the approved affiliation window. Uncertainty: delayed sponsor termination, duplicate external records, uncertain legal status, missing route ownership and controller propagation lag. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: hold the case, issue no recommendation to implement and route each gap to its authoritative owner.

ANSWER BOUNDARY

It must not perform identity proofing, retain identity-document contents, decide immigration or employment eligibility, or infer trustworthiness from affiliation. It must not decide identity, affiliation, role legitimacy, sponsorship or prerequisite outcomes; expose protected person or credential records beyond the authorised case; authorise or execute access changes; promise entry; or treat a transaction as proof of passage.

WHY THIS IS USEFUL

The administrator can create a sponsor-owned, time-bounded case instead of an indefinite external account.

AC-010

QUESTION 10 OF 10

STAKEHOLDER QUESTION

Which records in this new cohort are ready for batch preparation under current rules, and which must be held for individual review?

SENSORS AND TELEMETRY

Proposed evidence and location: batch request records, authoritative identity and affiliation feeds, current role-template versions, prerequisite status, duplicate detection and dry-run controller distribution results. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: every named cohort member, the current cohort role and dates, target groups, prerequisites, individual exceptions and all mapped controllers affected by the batch. Timing/aggregation: at batch import, after source or template updates, before hand-off and after authorised propagation; report counts by READY, HOLD, CONFLICT and exception type; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate file schema and identity matching with test rows, check duplicate and missing identifiers, verify template hashes and sample controller acknowledgements across the affected estate; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: batch completeness can conceal individual conflicts, stale copied exceptions, future-effective identities, duplicate rows and partial controller coverage.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the cohort owner or programme sponsor owns membership and purpose; HR or Student Records own affiliation; training and area owners own prerequisites; Security owns current templates and batch rules. Current authoritative membership and template versions outrank last year's file, copied groups, prior exceptions and observed attendance. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A feed is stale after its owner-approved freshness limit; if no limit is documented, more than two missed scheduled updates, a schema change, or an unexplained count discontinuity makes the affected result UNKNOWN. Show conflicting values side by side and route them to the cohort sponsor, identity, prerequisite or Security owner linked to each failed row; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: comparison, calculation and recommendation. Decision interval: at each batch preparation cycle and immediately before authorised implementation. Minimum evidence: stable identity IDs, current cohort membership and dates, approved role template, prerequisite records, exception decisions and complete affected-controller map. Method: validate each row independently, compare against the current template, quarantine duplicates and copied exceptions, then produce aggregate readiness counts and an itemised hold list. Uncertainty: identity mismatches, mixed effective dates, unmachine-readable approvals, partial asset mapping, offline controllers and exceptional cases hidden inside a large batch. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: implement nothing through the analytical service; mark unsupported rows HOLD or CONFLICTED and keep them out of the ready set.

ANSWER BOUNDARY

It must not treat batch membership as proof of eligibility, copy prior-year exceptions, suppress person-specific conflicts in an aggregate or issue credentials. It must not decide identity, affiliation, role legitimacy, sponsorship or prerequisite outcomes; expose protected person or credential records beyond the authorised case; authorise or execute access changes; promise entry; or treat a transaction as proof of passage.

WHY THIS IS USEFUL

The administrator can prepare a clean, auditable cohort batch while isolating the few cases that need owner attention.

Credential issue, replacement and lifecycle

AC-011 to AC-020

AC-011

QUESTION 1 OF 10

STAKEHOLDER QUESTION

Is this personalised credential ready for secure issue and collection, or should it remain on hold?

SENSORS AND TELEMETRY

Proposed evidence and location: credential-management workflow, secure stock custody scans, personalisation-station audit logs, encoder self-test status and approved test-reader results at the issue desk and credential item.

Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the specific credential serial or token, linked identity case, issue workstation, stock location, collection appointment and the approved test reader used for validation.

Timing/aggregation: at personalisation, after each custody hand-off, immediately before collection and at end-of-day reconciliation; show item status inside the authorised case and aggregate stock counts elsewhere; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: run the manufacturer-approved encoder test, use a designated non-person test credential and test reader, verify stock serial scans and align workstation, PACS and custody clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: successful encoding does not prove the collector's identity, physical hand-over or operation at every door; printer, stock and PACS records may update at different times.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Identity and Access Management owns the identity link; Security or the credential office owns issue policy and token status; the stock custodian owns physical custody; the authorised checker owns collection verification. A verified issue case, current credential status, accountable custody record and completed approved collection check outrank printer success, an appointment booking or verbal confirmation. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case.

Staleness and conflict: A custody or inventory record is stale when the latest expected reconciliation is missing, the item identifier cannot be matched, or a hand-off lacks accountable acknowledgement. Show conflicting values side by side and route them to the identity, credential-office, stock-custody or Security owner associated with the failed gate; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup and rules-based validation. Decision interval: after personalisation, before hand-over and after any custody discrepancy. Minimum evidence: stable identity and credential IDs, approved issue case, current status, successful controlled encoding test, custody location, collection method and required checker role. Method: evaluate issue-policy gates in order, reconcile stock and PACS status and distinguish READY, HOLD, CUSTODY CONFLICT and TECHNICAL RETEST. Uncertainty: unrecorded hand-offs, delayed stock scans, duplicate token bindings, test-reader coverage limits and human identity checks that occur outside the system. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: keep the credential in secure custody, mark it HOLD or UNKNOWN and route the exact failed gate to the responsible owner.

ANSWER BOUNDARY

It must not conduct the physical identity check, reveal credential data to an unauthorised collector, record secret encoding material or hand over the item. It must not encode, issue, replace, suspend, revoke or destroy a credential; handle PINs, keys or cryptographic secrets; disclose token or event data beyond the authorised case; promise entry; or infer the holder's identity, location or intent.

WHY THIS IS USEFUL

The administrator can issue only technically and procedurally ready credentials while keeping ambiguous items in accountable custody.

AC-012

QUESTION 2 OF 10

STAKEHOLDER QUESTION

After this lost or stolen credential report, which containment steps are confirmed and what must be resolved before replacement?

SENSORS AND TELEMETRY

Proposed evidence and location: incident case status, credential lifecycle state, suspension or revocation instruction records, distribution queues, controller acknowledgements and replacement workflow at the token and affected controller. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the reported token, any linked mobile or secondary credential, every controller that can enforce its groups and the replacement case; no access-event history is required for routine containment status. Timing/aggregation: from report receipt through each status transition and controller acknowledgement; display propagation by controller state and age, not a movement trail; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate token-to-identity binding, instruction identifiers and controller receipt with test cases; verify queue sequence, clock offsets and offline-controller last-contact time; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a central status change does not prove enforcement at offline controllers, and a report does not establish where the token is or whether it was misused.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the credential holder or authorised reporter owns the report; Security owns containment instruction and risk decision; the credential office owns replacement; PACS operations owns controller distribution status. A verified Security instruction and controller acknowledgement outrank a user report alone, a central-screen status without distribution evidence and assumptions based on missing events. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to Security, the credential office or PACS operations according to the unresolved stage; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, observation and propagation estimate. Decision interval: immediately on report, at each enforcement update and before replacement issue. Minimum evidence: verified token ID, authorised containment instruction, current central state, target controller list, acknowledgement or last-contact evidence and replacement approval status. Method: trace the instruction from case to central status to each controller, calculate confirmed, queued, offline and unknown coverage, and separate replacement readiness from containment. Uncertainty: offline panels, stale mappings, delayed queues, multiple active tokens and incomplete reports; estimate uncertainty grows with last-contact age. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: treat unacknowledged coverage as UNKNOWN, do not issue a replacement through the service and route residual exposure to Security.

ANSWER BOUNDARY

It must not decide whether the report is truthful, infer misuse or location, search movement history by default, or itself suspend, revoke or replace the credential. It must not encode, issue, replace, suspend, revoke or destroy a credential; handle PINs, keys or cryptographic secrets; disclose token or event data beyond the authorised case; promise entry; or infer the holder's identity, location or intent.

WHY THIS IS USEFUL

The administrator can show Security exactly where containment is confirmed and avoid treating replacement as proof that the old token is ineffective.

Credential issue, replacement and lifecycle

AC-011 to AC-020

AC-013

QUESTION 3 OF 10

STAKEHOLDER QUESTION

Is this failed access attempt attributable to the credential, reader or door path, schedule, or entitlement configuration?

SENSORS AND TELEMETRY

Proposed evidence and location: controlled diagnostic transactions using the affected credential at an approved test reader and a non-person test credential at the affected reader, plus reader diagnostics, deny reason, controller clock, schedule and group configuration. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the affected credential case, the reported reader and controller, one independent approved test reader, the relevant permission group and the exact failure window. Timing/aggregation: during a short authorised diagnostic window, with repeated tests separated from ordinary use; retain raw test events in the case and report only the diagnostic outcome elsewhere; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: verify the test credential, test reader and reader firmware against commissioning records, confirm door/reader mapping, align clocks and record environmental or hardware observations only when made by a competent technician; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a successful read is not proof of passage, intermittent faults may not reproduce, radio-frequency interference may be local, and deny codes can be vendor-specific or misconfigured.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the credential office owns token condition; PACS operations owns entitlement and reader configuration; Security Systems or an approved maintainer owns field diagnosis; the sponsor or area owner owns access eligibility. Controlled test evidence and current configuration outrank anecdotal reports, a single ordinary transaction and assumptions that a denial means entitlement failure. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to the credential office, PACS operations, area owner or field maintainer matching the isolated branch; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: diagnosis by controlled comparison. Decision interval: when a fault case is opened, after each controlled test and before deciding the service route. Minimum evidence: verified affected token, exact reader and time, current entitlement and schedule, controlled test credential, independent test reader and diagnostic codes. Method: use a decision tree: compare token performance across readers, reader performance across tokens, then entitlement, schedule and clock state; rank supported causes rather than assert one. Uncertainty: non-reproducible failure, reader sensitivity variation, stale deny-code mapping, local interference, clock drift and incomplete test coverage. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return UNKNOWN or a ranked candidate set; make no entitlement change and route to replacement, configuration or field service only when minimum evidence supports it.

ANSWER BOUNDARY

It must not infer that the credential holder attempted unauthorised entry, disclose their other transactions, or remotely tune hardware beyond approved maintenance procedure. It must not encode, issue, replace, suspend, revoke or destroy a credential; handle PINs, keys or cryptographic secrets; disclose token or event data beyond the authorised case; promise entry; or infer the holder's identity, location or intent.

WHY THIS IS USEFUL

The administrator can send the case to the right service path instead of replacing cards or changing permissions by guesswork.

AC-014

QUESTION 4 OF 10

STAKEHOLDER QUESTION

Which credentials will expire within the review horizon, and which have a current authorised basis for renewal?

SENSORS AND TELEMETRY

Proposed evidence and location: credential expiry index, affiliation and sponsor end dates, renewal workflow, prerequisite validity, lifecycle job status and controller distribution state. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: all credentials within the policy-defined planning horizon, linked identities and affiliations, renewal approvals, groups and controllers that would be affected by expiry or renewal. Timing/aggregation: daily or at the approved lifecycle batch interval, with alerts at policy-defined lead times; report aggregate counts by expiry window and authorised case detail only; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate date and time-zone handling with boundary test cases, reconcile token expiry against affiliation and approval dates, and verify lifecycle job and controller clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: credential expiry, affiliation end and entitlement end can be different; a future renewal request is not approval, and offline controllers may retain an older state.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the credential office owns token expiry policy; HR, Student Records or collaborator registry owns affiliation; sponsors and area owners own continuation; Security owns renewal rules. The earliest applicable authoritative end date and current approval basis outrank historical renewal, user expectation and a token's technical capacity to remain valid. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: An approval or case record is stale when expired, superseded, outside its effective window, or missing a current owner/version; a connector state beyond the owner-approved freshness limit is UNKNOWN. Show conflicting values side by side and route them to the affiliation, sponsor, area, credential-office or Security owner responsible for renewal evidence; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: forecast, lookup and recommendation. Decision interval: at each lifecycle run, when source dates change and before renewal implementation. Minimum evidence: current token and affiliation end dates, renewal request and approvals, prerequisite validity, applicable policy horizon and controller distribution map. Method: calculate time to each controlling end date, classify renew, lapse as planned, hold or conflict, and forecast the workload and potential service gap without extending anything. Uncertainty: date semantics, daylight-saving boundaries, pending approvals, delayed source feeds, duplicate credentials and offline enforcement. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: leave existing dates unchanged, mark unsupported renewals HOLD and escalate only cases inside the owner-defined lead window.

ANSWER BOUNDARY

It must not extend a credential, assume continued affiliation, promise uninterrupted access or treat past renewal as authority for the future. It must not encode, issue, replace, suspend, revoke or destroy a credential; handle PINs, keys or cryptographic secrets; disclose token or event data beyond the authorised case; promise entry; or infer the holder's identity, location or intent.

WHY THIS IS USEFUL

The administrator can stage a defensible renewal queue early while allowing genuinely ending credentials to lapse as authorised.

Credential issue, replacement and lifecycle

AC-011 to AC-020

AC-015

QUESTION 5 OF 10

STAKEHOLDER QUESTION

Does this identity have more than one active credential, and is each overlap justified, time-bounded and owner-approved?

SENSORS AND TELEMETRY

Proposed evidence and location: token-to-identity binding records, credential status, issue and return custody events, mobile-credential enrolment status and controller distribution evidence. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: all physical, mobile, temporary and replacement tokens linked to the same authoritative identity, including active windows, purpose, sponsor and affected groups. Timing/aggregation: on issue, replacement, return, mobile enrolment and periodic reconciliation; show token-count and overlap duration, not transaction history; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate identity linkage and token uniqueness with controlled duplicates, reconcile serials against inventory, and verify effective dates and status clocks across credential platforms; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: multiple tokens can be legitimate during controlled transition; duplicate identity records, delayed returns and offline controllers can make the central count incomplete.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Identity and Access Management owns identity linkage; the credential office owns token binding and custody; Security owns overlap policy and exceptions; sponsors own approved purpose. Current identity linkage, documented overlap purpose and time-bounded Security approval outrank assumptions that one token per person is always required or that every overlap is acceptable. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A custody or inventory record is stale when the latest expected reconciliation is missing, the item identifier cannot be matched, or a hand-off lacks accountable acknowledgement. Show conflicting values side by side and route them to Identity and Access Management, the credential office or Security; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup and comparison. Decision interval: on every lifecycle event and at the approved reconciliation interval. Minimum evidence: stable identity ID, complete token inventory across supported platforms, status and effective dates, custody state, purpose and any overlap approval. Method: group tokens by identity, calculate overlapping active windows, compare each overlap with policy and distinguish approved transition, unknown duplicate and stale return. Uncertainty: cross-platform linkage gaps, delayed status propagation, temporary credentials recorded outside the main system and legitimate exceptional overlap. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: do not disable any token; mark unexplained overlaps CONFLICTED and route them to the credential office and Security.

ANSWER BOUNDARY

It must not infer misuse, sharing or attendance from multiple credentials, or reveal token details outside the approved lifecycle case. It must not encode, issue, replace, suspend, revoke or destroy a credential; handle PINs, keys or cryptographic secrets; disclose token or event data beyond the authorised case; promise entry; or infer the holder's identity, location or intent.

WHY THIS IS USEFUL

The administrator can distinguish approved transition windows from unexplained duplicates before issuing another credential.

AC-016

QUESTION 6 OF 10

STAKEHOLDER QUESTION

Which personalised credentials remain uncollected beyond policy, and who currently has documented custody of each?

SENSORS AND TELEMETRY

Proposed evidence and location: personalisation completion records, secure cabinet or envelope custody scans, collection appointment status, authorised hand-off records and stock reconciliation at the issue location. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: every personalised but not completed issue case, its credential serial, current secure location, accountable custodian and age since ready-for-collection. Timing/aggregation: at each custody hand-off and daily or at the policy-defined reconciliation interval; report age bands and custody exceptions, with item detail only to authorised staff; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: verify scanner and workstation identifiers, reconcile serials by controlled stock counts, test custody workflow transitions and align appointment, personalisation and scan clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: missing collection does not prove non-attendance or abandonment; manual cabinet moves, delayed scans and cancelled identities can leave custody records behind reality.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the credential office owns collection policy; the stock custodian owns physical location; Identity, HR or Student Records owns identity validity; Security owns disposal or escalation rules. Current physical custody confirmation and credential status outrank an appointment record, printer output or an assumption based on elapsed time alone. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A custody or inventory record is stale when the latest expected reconciliation is missing, the item identifier cannot be matched, or a hand-off lacks accountable acknowledgement. Show conflicting values side by side and route them to the current stock custodian, credential office or identity owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: calculation and reconciliation. Decision interval: daily, at custody change and before any disposal or reissue decision. Minimum evidence: credential serial, personalisation time, current custody location and custodian, collection status, identity validity and applicable age threshold. Method: calculate age from the verified ready time, reconcile every item to one physical custody state and classify follow-up, revalidation or discrepancy without contacting the person automatically. Uncertainty: late scans, shared storage, cancelled appointments, identity end dates and unrecorded hand-offs. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: keep the item secured, mark custody UNKNOWN where unreconciled and require a witnessed check before any further lifecycle action.

ANSWER BOUNDARY

It must not infer why a credential was not collected, disclose the person's case to unauthorised staff, dispose of stock or contact the person without approved procedure. It must not encode, issue, replace, suspend, revoke or destroy a credential; handle PINs, keys or cryptographic secrets; disclose token or event data beyond the authorised case; promise entry; or infer the holder's identity, location or intent.

WHY THIS IS USEFUL

The administrator can direct a precise custody check and prevent uncollected personalised items from becoming unowned stock.

Credential issue, replacement and lifecycle

AC-011 to AC-020

AC-017

QUESTION 7 OF 10

STAKEHOLDER QUESTION

Which loan, visitor or contractor credentials are overdue, and which sponsor or custodian owns recovery action?

SENSORS AND TELEMETRY

Proposed evidence and location: loan issue and return custody scans, visitor or contractor case status, badge pool inventory and sponsor acknowledgement at reception, security desk or approved issue point. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: all reusable loan credentials currently issued, their case, sponsor, planned return time, current status and the badge pool from which they were taken. Timing/aggregation: at issue, planned return, actual return and end-of-shift reconciliation; report counts by age, sponsor and pool, not door-event activity; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate badge serial scans and pool membership, test issue/return workflow transitions, reconcile physical counts and align visitor, reception and PACS clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: an overdue custody record does not reveal who holds the badge or where it is; sponsor changes, emergency retention and delayed reception closure can be legitimate.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: reception or Security owns loan custody; the sponsor owns the visitor or work purpose; contractor management owns contractor return; Security owns overdue and containment policy. The signed or otherwise accountable issue record and current physical reconciliation outrank assumptions from access events, booking status or last-known desk location. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A custody or inventory record is stale when the latest expected reconciliation is missing, the item identifier cannot be matched, or a hand-off lacks accountable acknowledgement. Show conflicting values side by side and route them to the recorded sponsor, current custodian or Security; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup and calculation. Decision interval: at the planned return time, at shift reconciliation and before badge-pool reuse. Minimum evidence: badge serial, issue case, accountable sponsor or custodian, planned return, current status, pool inventory and any authorised extension. Method: calculate overdue duration, check approved extension or incident status, identify the accountable follow-up owner and keep the badge state separate from bearer location. Uncertainty: missing scans, pooled badges, after-hours returns, sponsor changes, emergency use and partial inventory counts. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: mark the item LOCATION UNKNOWN, prevent analytical assumptions and route follow-up through the recorded owner under approved procedure.

ANSWER BOUNDARY

It must not search movement history to locate the bearer by default, accuse a person of retention, contact them outside approved procedure or remotely alter the badge. It must not encode, issue, replace, suspend, revoke or destroy a credential; handle PINs, keys or cryptographic secrets; disclose token or event data beyond the authorised case; promise entry; or infer the holder's identity, location or intent.

WHY THIS IS USEFUL

The administrator can focus follow-up on the accountable sponsor or custodian and protect the integrity of the reusable badge pool.

AC-018

QUESTION 8 OF 10

STAKEHOLDER QUESTION

Is this mobile credential enrolment complete, privacy-compliant and supported across the commissioned reader estate?

SENSORS AND TELEMETRY

Proposed evidence and location: provisioning transaction status, non-secret device-eligibility attestation, wallet or app enrolment outcome, reader compatibility matrix, controlled test transactions and privacy approval state. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the authorised enrolment case, supported device class and operating environment, targeted reader models and firmware, and the groups intended for the mobile token. Timing/aggregation: at enrolment, after app, operating-system or reader changes and before broad deployment; retain person-level enrolment detail only in the case and report compatibility by device/reader class; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: use approved non-person test devices and credentials, validate reader firmware and antenna configuration, verify signed compatibility records and align app, platform, reader and PACS clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: successful provisioning does not prove dependable operation across all devices or buildings; device state can change, radio conditions vary and vendor status may be opaque.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the mobile-credential service owner owns provisioning; Information Security owns security acceptance; Data Protection owns lawful design; Accessibility owns inclusive use; Security Systems owns reader compatibility. Approved architecture, DPIA or equivalent decision, current compatibility test evidence and authorised enrolment outrank vendor claims, a single successful phone and informal pilot experience. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to the mobile service, Information Security, Data Protection, Accessibility or Security Systems owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, comparison and recommendation. Decision interval: at enrolment, on relevant technology change and before deployment expansion. Minimum evidence: authorised identity case, approved device eligibility, current privacy and security decisions, supported reader matrix, successful controlled test and fallback method. Method: compare the enrolment and target estate with the approved compatibility matrix, separate technical, governance and accessibility gates and identify unsupported combinations. Uncertainty: vendor telemetry gaps, device changes, reader drift, inconsistent radio performance, outdated test coverage and inaccessible fallback arrangements. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: do not provision or expand through the service; return NOT READY for any failed gate and preserve an approved alternative credential route.

ANSWER BOUNDARY

It must not access device contents, collect unnecessary device identifiers, handle cryptographic keys, force mobile use or certify the technology secure or accessible. It must not encode, issue, replace, suspend, revoke or destroy a credential; handle PINs, keys or cryptographic secrets; disclose token or event data beyond the authorised case; promise entry; or infer the holder's identity, location or intent.

WHY THIS IS USEFUL

The administrator can hold unsupported enrolments and deploy only where technical, privacy and accessibility evidence align.

Credential issue, replacement and lifecycle

AC-011 to AC-020

AC-019

QUESTION 9 OF 10

STAKEHOLDER QUESTION

What credential transition from temporary to continuing status avoids duplicate active tokens and a service gap?

SENSORS AND TELEMETRY

Proposed evidence and location: old and proposed credential lifecycle states, affiliation transition records, entitlement snapshots, issue and return custody events, overlap approval and controller acknowledgements. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the verified identity, temporary and continuing credentials, all linked groups, the approved transition window and every controller that can enforce either token. Timing/aggregation: at status change, before new issue, at planned hand-off and after propagation; show a transition timeline and overlap status rather than access events; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate token bindings and effective dates, test transition rules with synthetic cases, reconcile custody and verify controller receipt of the authorised state; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: the new affiliation may become effective before collection, the old token may remain in custody, and offline controllers may enforce an earlier state.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: HR, Student Records or collaborator registry owns the status transition; the credential office owns issue and return; Security owns permitted overlap and entitlement continuity. The authoritative effective status and approved transition plan outrank a technical ability to keep both tokens active, previous practice and assumed collection dates. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: An approval or case record is stale when expired, superseded, outside its effective window, or missing a current owner/version; a connector state beyond the owner-approved freshness limit is UNKNOWN. Show conflicting values side by side and route them to the affiliation owner, credential office or Security; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: temporal comparison and recommendation. Decision interval: on transition approval, before issue and at the end of any overlap window. Minimum evidence: verified identity, old and new status dates, both token states, custody plan, approved overlap duration, entitlement mapping and controller coverage. Method: construct the authorised timeline, identify gap and overlap intervals, compare token scopes and recommend the shortest supported transition sequence without executing it. Uncertainty: future-effective dates, delayed collection, offline controllers, mismatched group scopes and unrecorded temporary extensions. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: keep the existing state, mark the transition HOLD where continuity and duplication cannot both be bounded, and route for owner decision.

ANSWER BOUNDARY

It must not decide which token to disable, assume collection occurred, infer use from events or extend the temporary credential to avoid workflow delay. It must not encode, issue, replace, suspend, revoke or destroy a credential; handle PINs, keys or cryptographic secrets; disclose token or event data beyond the authorised case; promise entry; or infer the holder's identity, location or intent.

WHY THIS IS USEFUL

The administrator can stage a controlled credential hand-over that avoids both an access gap and an indefinite duplicate token.

AC-020

QUESTION 10 OF 10

STAKEHOLDER QUESTION

Do issued credentials, secure stock, returned items, quarantined items and witnessed destruction records reconcile?

SENSORS AND TELEMETRY

Proposed evidence and location: credential inventory exports, secure stock scans, issue and return registers, encoder output logs, witnessed destruction records and lifecycle status across each approved storage and issue point. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: all credential serial ranges and token types in scope, their latest custody or lifecycle state, accountable location and any batch or destruction certificate. Timing/aggregation: at each hand-off and destruction event, with full reconciliation at the policy-defined interval; report counts by state and itemised discrepancies to authorised custodians only; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: perform controlled physical counts, verify scanner identifiers and serial formats, reconcile batch totals, test duplicate and missing-item detection and align custody-system clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: systems can use different serial representations, stock may be in transit, scans may be delayed and destruction evidence may exist outside the main credential system.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the stock custodian owns physical inventory; the credential office owns lifecycle records; authorised witnesses own destruction confirmation; Security and Records Management own reconciliation policy. Current witnessed physical evidence and accountable custody records outrank system counts alone, printer output, assumed destruction or undocumented transfer. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A custody or inventory record is stale when the latest expected reconciliation is missing, the item identifier cannot be matched, or a hand-off lacks accountable acknowledgement. Show conflicting values side by side and route them to the stock custodian, credential office, authorised witness or Security; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: calculation and reconciliation. Decision interval: after material custody events and at every scheduled inventory control. Minimum evidence: complete serial list, current physical count, issue/return records, destruction certificates, transfer acknowledgements and defined reconciliation scope. Method: normalise serial formats, construct a one-state-per-item ledger, compare totals and item paths, and classify timing difference, data error or unexplained discrepancy. Uncertainty: in-transit stock, delayed scans, partial counts, duplicate serial entry, external destruction records and uncertain cut-off time. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: quarantine unexplained items or batches under existing procedure, make no lifecycle change and require witnessed reconciliation.

ANSWER BOUNDARY

It must not declare a loss, accuse a custodian, destroy or reissue stock, or expose serial-level data beyond authorised inventory staff. It must not encode, issue, replace, suspend, revoke or destroy a credential; handle PINs, keys or cryptographic secrets; disclose token or event data beyond the authorised case; promise entry; or infer the holder's identity, location or intent.

WHY THIS IS USEFUL

The administrator can reconcile custody and lifecycle records before discrepancies become credential-control failures.

Doors, zones, permission groups and dependencies

AC-021 to AC-030

AC-021

QUESTION 1 OF 10

STAKEHOLDER QUESTION

Which active permission group currently controls this opening, and through which direct or inherited paths?

SENSORS AND TELEMETRY

Proposed evidence and location: versioned PACS configuration snapshots linking reader, door interface, controller, access level, nested group and schedule, with optional controlled test transaction at the named opening. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the exact commissioned opening and reader, every direct and inherited group path that can authorise it, the active schedule rules and the controller copy currently in force. Timing/aggregation: on request, after configuration change and after controller acknowledgement; show the configuration path and version, not person-level use; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: verify the physical asset label against the controller channel, use a designated test credential when a live check is authorised, compare configuration hashes and align controller and server clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a configured permission path does not prove physical entry, door availability or current room use; shared, nested or local controller rules may be absent from a central summary.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Security Systems owns PACS configuration; Estates owns the current door asset identifier; the area owner owns destination purpose; lift or specialist-system owners own their dependent mappings. The commissioned asset-to-controller mapping and current controller configuration outrank display labels, room names, observed use and the May 2021 floor plan. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to Security Systems and the current Estates or area asset owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup and graph traversal. Decision interval: on operational query and after any relevant configuration or asset change. Minimum evidence: verified opening and reader IDs, current controller configuration hash, access-level and nested-group graph, schedule state and asset mapping. Method: trace every authorising path from the opening back through schedules and groups, distinguish direct, inherited and local-controller rules and present the exact version. Uncertainty: stale controller copies, duplicate labels, unmapped channels, hidden local logic, scheduled modes and configuration replication delay. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return UNKNOWN for any unverified path and require physical or configuration reconciliation before relying on the result.

ANSWER BOUNDARY

It must not identify who used the opening, infer passage from a transaction, decide whether a group should grant entry or change the configuration. It must not alter groups or controller configuration; operate doors, lifts or emergency modes; certify safety, security or compliance; disclose identifiable transaction histories; or infer physical passage, occupancy or intent.

WHY THIS IS USEFUL

The administrator can find the real permission path before editing the wrong group or giving an inaccurate explanation.

AC-022

QUESTION 2 OF 10

STAKEHOLDER QUESTION

Will the proposed destination permission also supply every approved upstream door, gate and lift-floor dependency?

SENSORS AND TELEMETRY

Proposed evidence and location: current group-to-opening graph, route and lift-floor dependency mappings, schedule rules and dry-run entitlement results; no occupancy or route-tracking sensor is required. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the approved origin class, destination, accessible alternatives, controlled openings and lift floors that form each owner-approved route, including shared entrances and time profiles. Timing/aggregation: during request design and after any route, lift, closure or group change; summarise covered, missing and excess dependencies by route option; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate graph edges against commissioned assets and current route-owner records, test representative routes with non-person credentials and verify lift-interface and controller versions; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a graph cannot establish that a route is currently open, safe, accessible or unobstructed; historical plans and access events cannot establish current routing.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the destination area owner owns need; Estates and Accessibility owners own current route records; lift specialists own floor-control mapping; Security owns access groups and schedules. Current owner-approved route and asset records, plus commissioned access mappings, outrank the May 2021 floor plan, shortest-path calculation and prior user movement. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to the route, Accessibility, lift, area or Security owner responsible for the missing dependency; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, graph reachability and recommendation. Decision interval: when designing or reviewing the access package and after a relevant route change. Minimum evidence: approved destination and origin assumptions, current route options, asset mappings, lift-floor dependencies, schedule rules and current closure information. Method: calculate access reachability for each approved route, separate missing from excess permissions and recommend the smallest supported dependency set without certifying the route. Uncertainty: unmapped doors, temporary closures, shared routes, lift operating modes, accessibility constraints and controller copies that have not acknowledged change. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return UNKNOWN where a current route or dependency is not owner-verified; do not broaden access as a fallback.

ANSWER BOUNDARY

It must not choose a route for a person, certify accessibility or safety, infer where anyone travelled, or treat access permission as evidence that the route is usable. It must not alter groups or controller configuration; operate doors, lifts or emergency modes; certify safety, security or compliance; disclose identifiable transaction histories; or infer physical passage, occupancy or intent.

WHY THIS IS USEFUL

The administrator can avoid issuing a destination group that fails at an upstream door or grants unnecessary alternative areas.

Doors, zones, permission groups and dependencies

AC-021 to AC-030

AC-023

QUESTION 3 OF 10

STAKEHOLDER QUESTION

After building works, does the PACS record for this opening still match the current door asset, zone boundary and approved use?

SENSORS AND TELEMETRY

Proposed evidence and location: asset-label and controller-channel records, commissioned reader identity, door-position contact open/closed test, lock-monitor and request-to-exit test where fitted, plus as-built handover evidence at the opening. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the single opening and its reader, lock, contact, controller channel, zone and current asset record, together with every PACS object that uses the mapping. Timing/aggregation: at commissioning, after building works, hardware replacement or room reallocation and before configuration change; retain test observations and configuration snapshots with exact times; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: perform witnessed open/closed and REX tests, use a designated test credential, verify asset labels physically, compare as-built and controller identifiers and use calibrated technician instruments where electrical measurement is needed; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a historical floor plan cannot establish current use or mapping; some contacts may not be fitted, wiring labels may be wrong and a passed test covers only the observed state and time.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Estates owns the current asset register and room-use record; Security Systems owns PACS mapping; the project or maintenance handover owner owns as-built evidence; the area owner confirms operational purpose. Physically verified commissioned identifiers and current owner records outrank the May 2021 floor plan, legacy labels, previous configuration and inferred room use. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to Estates, the project handover owner, Security Systems and the area owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: comparison and diagnosis. Decision interval: after relevant works and before the opening or associated group is accepted for service. Minimum evidence: physical asset and reader IDs, controller channel, witnessed functional tests, current room/zone record, as-built evidence and impacted PACS object list. Method: compare physical, asset-register, as-built and PACS mappings, identify exact mismatches and calculate the configuration objects that depend on each disputed identifier. Uncertainty: missing handover records, hidden wiring changes, partial tests, absent sensors, stale labels and room-use decisions not yet reflected in systems. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: mark the opening UNVERIFIED, do not use its mapping for entitlement design and require competent physical reconciliation.

ANSWER BOUNDARY

It must not determine current room use from the floor plan, certify electrical or fire safety, relabel assets or alter controller channels. It must not alter groups or controller configuration; operate doors, lifts or emergency modes; certify safety, security or compliance; disclose identifiable transaction histories; or infer physical passage, occupancy or intent.

WHY THIS IS USEFUL

The administrator can stop configuration work from targeting the wrong physical opening after refurbishment or re-cabling.

AC-024

QUESTION 4 OF 10

STAKEHOLDER QUESTION

What documented fail-safe or fail-secure, power-loss, fire-interface and manual-override arrangement applies to this opening?

SENSORS AND TELEMETRY

Proposed evidence and location: approved configuration records, supervised lock and door-contact states where fitted, power-supply or UPS health, fire-interface monitored status and current competent test certificates at the opening and its controller. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the named opening, locking hardware, local power path, controller input/output, fire or emergency interface and any approved mechanical or manual override. Timing/aggregation: live health at the configured heartbeat and documentary lookup on demand; functional tests only under an authorised plan and competent supervision, never by routine analytics; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: verify sensor input states against witnessed physical tests, validate PSU telemetry against calibrated technician measurements, confirm interface mapping and align all test and event clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: telemetry cannot prove safe behaviour under every fault; not all hardware exposes monitored state, and configuration text may not match wiring or current emergency procedure.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: competent fire, electrical and security professionals own safety-critical design and test; Estates owns asset and maintenance records; Security Systems owns PACS configuration; incident procedures own emergency operation. Current signed design, as-built and competent test evidence outrank PACS labels, inferred hardware behaviour, past events and historical plans. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to the competent fire, electrical, security or Estates owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup and comparison. Decision interval: during design review, commissioning, relevant change and authorised fault investigation. Minimum evidence: verified opening and hardware IDs, current design intent, as-built wiring, configuration version, competent test result, maintenance status and emergency procedure reference. Method: assemble the evidence chain, compare documented design with commissioned configuration and observed test state, and identify gaps without predicting life-safety performance. Uncertainty: unmonitored components, expired tests, hidden wiring, local overrides, battery condition, interface latency and operating modes not exercised in the latest test. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return UNKNOWN and route to competent professionals; no remote test, release or change is initiated.

ANSWER BOUNDARY

It must not determine fail-safe policy, certify the opening, simulate a fire release, advise emergency command or operate an override. It must not alter groups or controller configuration; operate doors, lifts or emergency modes; certify safety, security or compliance; disclose identifiable transaction histories; or infer physical passage, occupancy or intent.

WHY THIS IS USEFUL

The administrator can provide a precise evidence pack and avoid treating a PACS setting as proof of safety-critical behaviour.

Doors, zones, permission groups and dependencies

AC-021 to AC-030

AC-025

QUESTION 5 OF 10

STAKEHOLDER QUESTION

Which time profiles, global modes, nested groups or controller rules can override this opening's ordinary permission decision?

SENSORS AND TELEMETRY

Proposed evidence and location: versioned configuration graph for access levels, nested groups, time profiles, global modes, holidays and controller-local rules, plus current mode and controller configuration hash. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the named opening and all upstream policy objects, schedules and local rules capable of changing its normal grant or deny result. Timing/aggregation: on query, before change and after controller acknowledgement; report active and future-effective override paths separately; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate graph extraction with vendor-supported test configurations, compare server and controller hashes and verify mode and schedule clocks against the approved time source; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: undocumented local logic, integration-driven modes, stale controller copies and vendor-specific rule order can make a central graph incomplete.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Security owns access policy and global modes; PACS operations owns configuration; building or incident authorities own approved exceptional operating modes; system vendors document rule precedence. Current approved policy and the configuration actually acknowledged by the controller outrank user-interface labels, expected rule order and historic operating practice. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to Security, PACS operations or the authority responsible for the exceptional mode; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, graph traversal and comparison. Decision interval: before entitlement or schedule change, during diagnosis and after deployment. Minimum evidence: verified opening, current server and controller configuration, rule-precedence documentation, schedule and mode state and integration inputs. Method: enumerate all authorising and denying paths in evaluated order, distinguish current from future conditions and compare central intent with local enforcement state. Uncertainty: hidden vendor rules, asynchronous mode changes, daylight-saving configuration, stale local cache and unmapped integration inputs. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return PARTIALLY VERIFIED or UNKNOWN where rule order or local state cannot be reproduced; make no configuration recommendation that assumes completeness.

ANSWER BOUNDARY

It must not activate a mode, decide emergency operation, infer why a mode was set or change a rule to force a desired outcome. It must not alter groups or controller configuration; operate doors, lifts or emergency modes; certify safety, security or compliance; disclose identifiable transaction histories; or infer physical passage, occupancy or intent.

WHY THIS IS USEFUL

The administrator can understand hidden dependencies before altering a group that appears simpler than the actual controller logic.

AC-026

QUESTION 6 OF 10

STAKEHOLDER QUESTION

What interlock, mantrap or anti-passback dependencies must be considered before changing this entitlement?

SENSORS AND TELEMETRY

Proposed evidence and location: controller state-machine configuration, reader direction mapping, door-position and lock-monitor states, request-to-exit inputs and controlled test sequences at every opening in the dependency. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the complete interlock or anti-passback set, including all participating doors, readers, controller channels, schedules, reset rules and integration states. Timing/aggregation: configuration lookup on demand and controlled sequence testing after relevant change; aggregate operational alarms by asset and interval rather than person; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: witness door-state transitions, validate reader direction and REX mapping, use non-person test credentials, verify state-reset behaviour and align controller clocks within the documented tolerance; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: events do not prove passage, tailgating or intent; missing REX or door contacts, manual overrides and out-of-sequence use can leave system state different from physical reality.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Security owns anti-passback policy; Security Systems owns controller logic; Estates owns door hardware; competent safety professionals own any life-safety constraints; area owners own operational purpose. Current commissioned state-machine design and witnessed test evidence outrank inferred sequence from access events, user reports and historic door arrangements. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to Security, Security Systems, Estates or the competent safety owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, state-machine comparison and diagnosis. Decision interval: before entitlement or rule change, after maintenance and during an authorised fault case. Minimum evidence: complete dependency membership, current logic version, reader direction, input mapping, reset rules, approved test plan and witnessed results. Method: model expected states, compare controlled test transitions with configured logic and identify unsupported or conflicting states without attributing them to people. Uncertainty: unobserved passage, missing sensors, manual resets, door bounce, clock skew, local overrides and incomplete test coverage. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: treat the state as UNKNOWN, avoid entitlement change and require controlled reset or field validation under approved procedure.

ANSWER BOUNDARY

It must not infer tailgating, occupancy or misconduct, operate an interlock, reset anti-passback state or certify the arrangement safe. It must not alter groups or controller configuration; operate doors, lifts or emergency modes; certify safety, security or compliance; disclose identifiable transaction histories; or infer physical passage, occupancy or intent.

WHY THIS IS USEFUL

The administrator can prevent an entitlement change from breaking a multi-door rule or leaving users trapped by an unrecognised dependency.

Doors, zones, permission groups and dependencies

AC-021 to AC-030

AC-027

QUESTION 7 OF 10

STAKEHOLDER QUESTION

Which lift car, landing and destination-floor permissions are required for the approved route to this destination?

SENSORS AND TELEMETRY

Proposed evidence and location: PACS-to-lift permission mapping, lift controller acknowledgements, landing reader configuration, destination-floor group graph and controlled functional tests by authorised lift and security specialists. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the requested destination, approved route, relevant lift cars and landings, floor-selection permissions, time profiles and any alternative accessible route. Timing/aggregation: during access design, after lift or access configuration change and at commissioned test; show dependency status by lift/floor, not passenger journeys; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: verify reader and lift-controller identifiers, use approved test credentials, witness floor-selection results, compare interface versions and align PACS and lift clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: permission does not establish lift availability, accessibility, current service state or safe route; lift modes and outages may change independently of PACS.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the lift service owner and competent lift contractor own lift function; Accessibility and Estates own current route records; Security owns access permissions; the destination owner owns need. Current lift-owner mapping and competent functional test evidence outrank floor-plan assumptions, reader labels, past successful use and central PACS configuration alone. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to the lift, Accessibility, Estates, destination or Security owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, graph reachability and comparison. Decision interval: during request design, commissioning and after relevant change. Minimum evidence: approved destination and route, current lift-floor mapping, access groups and schedules, controller acknowledgements, service status and authorised test result. Method: trace the complete door-and-lift permission path, compare intended and tested floor access and identify missing or excess permissions separately. Uncertainty: lift operating mode, service outage, inaccessible alternative, interface delay, local cache and incomplete test coverage across cars or floors. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return UNKNOWN for any unverified lift dependency and do not substitute broad floor access or claim the route is usable.

ANSWER BOUNDARY

It must not control a lift, certify lift safety or accessibility, infer passenger movement or use access data to determine occupancy. It must not alter groups or controller configuration; operate doors, lifts or emergency modes; certify safety, security or compliance; disclose identifiable transaction histories; or infer physical passage, occupancy or intent.

WHY THIS IS USEFUL

The administrator can include the exact lift-floor dependency instead of issuing a door group that still leaves the destination unreachable.

AC-028

QUESTION 8 OF 10

STAKEHOLDER QUESTION

Can access to the shared entrance be separated from the restricted internal zone without granting unnecessary building-wide rights?

SENSORS AND TELEMETRY

Proposed evidence and location: current door and group graph, nested permission structure, schedule rules and dry-run assignment simulation for the shared entrance and internal boundary; no occupancy sensing is required. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the shared entrance, all paths to the approved destination, restricted internal openings, lift or gate dependencies and every candidate group that touches them. Timing/aggregation: during permission design and after mapping or group change; report the exact required, excess and inseparable permission sets; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate opening-to-group mapping against commissioned identifiers, test graph calculations with known narrow and broad cases and verify controller configuration versions; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a technically separable group may not reflect current operational policy, and a graph cannot establish present route safety, opening hours or room use.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the destination and building owners own the business and zoning decision; Security owns group design; Estates and Accessibility own current route and asset records. Owner-approved zoning and current commissioned mapping outrank convenience, shortest-path logic, observed access patterns and historical plans. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to the building, destination, route or Security owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: set comparison and bounded recommendation. Decision interval: when designing or reviewing a permission package. Minimum evidence: approved destination and route, current group graph, opening mappings, schedules, shared dependencies and explicit restricted-zone exclusions. Method: calculate whether a group or combination covers the required path while excluding unrelated internal zones; quantify inseparable excess privilege. Uncertainty: shared controller rules, nested groups, unverified internal boundaries, temporary routing and business policy that has not been translated into access objects. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: do not recommend broad access where separation cannot be demonstrated; return the exact design gap for owner action.

ANSWER BOUNDARY

It must not decide building zoning, create a new group, certify a route or infer that low use makes excess access acceptable. It must not alter groups or controller configuration; operate doors, lifts or emergency modes; certify safety, security or compliance; disclose identifiable transaction histories; or infer physical passage, occupancy or intent.

WHY THIS IS USEFUL

The administrator can propose a narrow entrance-only design or show why a policy or configuration redesign is needed.

Doors, zones, permission groups and dependencies

AC-021 to AC-030

AC-029

QUESTION 9 OF 10

STAKEHOLDER QUESTION

Is the fire-alarm or emergency-release interface for this opening correctly mapped, and what current competent test evidence supports it?

SENSORS AND TELEMETRY

Proposed evidence and location: approved interface I/O mapping, monitored input and output state, controller configuration, event timestamps and current competent fire-interface test records; routine analytics does not initiate a test. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: every access-controlled opening and controller included in the defined fire or emergency release interface, including power, relay, monitoring and manual procedure dependencies. Timing/aggregation: documentary status on demand, live health at configured heartbeat and functional evidence only from authorised competent tests after commissioning or change; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: verify I/O labels physically, witness state changes under the approved test plan, compare controller and fire-system times, use calibrated electrical instruments where required and preserve signed test provenance; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a monitored state cannot prove correct release under all conditions; some interfaces may be hard-wired without rich telemetry, and central records may not match as-built wiring.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the competent fire-safety authority owns release requirements and test acceptance; Estates and electrical owners own wiring and power; Security Systems owns PACS mapping; incident command owns live emergency use. Current signed design, as-built documentation and competent test acceptance outrank PACS labels, historic alarm events, simulated inference and the May 2021 floor plan. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to the competent fire, electrical, Estates and Security Systems owners; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup and evidence comparison. Decision interval: at commissioning, after relevant change, during scheduled assurance and when an interface fault is suspected. Minimum evidence: complete interface asset list, current design and as-built mapping, PACS configuration version, competent test date/result, open defects and maintenance status. Method: reconcile design, as-built, configuration and test evidence by interface point, identify untested or mismatched elements and state the limits of observed coverage. Uncertainty: unmonitored wiring, partial tests, stale certificates, local bypasses, clock mismatch, undocumented change and test conditions that differ from a real incident. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return UNKNOWN for every unverified element and route to competent professionals; no release, alarm or mode command is generated.

ANSWER BOUNDARY

It must not test or operate the interface, certify fire compliance, predict emergency behaviour or override incident command and approved procedures. It must not alter groups or controller configuration; operate doors, lifts or emergency modes; certify safety, security or compliance; disclose identifiable transaction histories; or infer physical passage, occupancy or intent.

WHY THIS IS USEFUL

The administrator can supply a bounded assurance gap list instead of assuming that a configuration label proves emergency release readiness.

AC-030

QUESTION 10 OF 10

STAKEHOLDER QUESTION

What is the verified impact boundary of changing this access level across people, groups, schedules, doors and controllers?

SENSORS AND TELEMETRY

Proposed evidence and location: versioned configuration dependency graph, current assignment counts, effective schedules, controller distribution map and dry-run change simulation; person identities remain hidden unless the authorised change case requires them. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the proposed access level or group, every direct and nested assignment, controlled opening, lift floor, schedule, controller, integration and exception linked to it. Timing/aggregation: at change planning, after source changes and immediately before authorised implementation; report aggregate impact first and case-authorised item detail second; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate graph completeness against configuration inventory and commissioned assets, test impact calculations with synthetic nested changes and compare server and controller versions; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: off-workflow local rules, stale controller copies, unmapped integrations and future-effective assignments can be absent from the apparent impact set.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Security owns policy and change approval; PACS operations owns configuration; area, lift, fire and Estates owners own affected operational or safety dependencies; Identity owners own assignments. The approved change scope and current acknowledged configuration outrank assumptions based on group name, assignment count, historical room use or a central-only snapshot. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to Security and every named owner of an affected dependency; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: calculation, comparison and recommendation. Decision interval: during change design, at approval and immediately before implementation. Minimum evidence: current configuration graph, complete assignment and asset map, active and future schedules, controller versions, integration dependencies, approved change and rollback owner. Method: compute direct and transitive impact, separate current from future-effective effects, identify owners and produce a pre-change/rollback evidence package without executing the change. Uncertainty: hidden local logic, stale mappings, large nested groups, privacy-limited identity detail, controller replication delay and external integrations outside PACS visibility. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: mark the impact PARTIALLY VERIFIED or UNKNOWN and block analytical recommendation to proceed until critical dependencies are reconciled.

ANSWER BOUNDARY

It must not approve or implement the change, expose affected identities beyond authorised need, infer operational importance from usage, or promise rollback success. It must not alter groups or controller configuration; operate doors, lifts or emergency modes; certify safety, security or compliance; disclose identifiable transaction histories; or infer physical passage, occupancy or intent.

WHY THIS IS USEFUL

The administrator can plan owner consultation, testing and rollback around the real dependency footprint rather than the group label alone.

Schedules, temporary access and visitors

AC-031 to AC-040

AC-031

QUESTION 1 OF 10

STAKEHOLDER QUESTION

Which recurring time profile fits this approved out-of-hours duty without creating all-night or all-week access?

SENSORS AND TELEMETRY

Proposed evidence and location: approved duty-schedule records, current PACS time-profile configuration, holiday calendars, group assignments and controller clock status; no physical sensing is required. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the named duty, approved days and time windows, required openings and route dependencies, applicable holidays and every controller enforcing the profile. Timing/aggregation: during request design, after duty or calendar change and before implementation; express windows in the authoritative local time zone and show weekly totals and uncovered/excess intervals; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: test boundary times, overnight spans, daylight-saving transitions and holiday overrides with synthetic cases; verify controller time zone, NTP status and configuration hash; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: work may overrun, duty descriptions may omit setup or handover time, and a schedule cannot establish whether the building is open, staffed, safe or accessible.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the manager or sponsor owns duty timing; building and Security owners own out-of-hours policy; Estates owns current opening or closure status; PACS operations owns time-profile implementation. Explicit approved duty windows and current building/security rules outrank broad role norms, past attendance, calendar assumptions and historical opening patterns. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to the duty, building, Security or Estates owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: calculation, comparison and recommendation. Decision interval: when the recurring access is designed, reviewed or materially changed. Minimum evidence: approved duty days and times, effective dates, required route, current holidays and closures, applicable out-of-hours rules and controller clock evidence. Method: construct the narrowest recurring intervals that cover the approved duty plus documented setup or handover margins, then quantify excess and conflicts. Uncertainty: ambiguous margins, overnight intervals, daylight-saving change, unverified building hours, holiday overrides and controller clock drift. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return no profile where the approved window or current operating conditions are unclear; keep access unchanged and route the conflict.

ANSWER BOUNDARY

It must not infer working hours from past transactions, decide that lone or out-of-hours work is safe, authorise overtime or create the time profile. It must not approve a visitor, contractor, event or working arrangement; create or activate schedules; infer attendance or route; promise entry; or override Security, sponsor, area-owner or incident decisions.

WHY THIS IS USEFUL

The administrator can propose a narrow recurring profile instead of defaulting to continuous access.

AC-032

QUESTION 2 OF 10

STAKEHOLDER QUESTION

Is this one-off permission limited to the exact openings and authorised start and end times, including controller propagation margins?

SENSORS AND TELEMETRY

Proposed evidence and location: temporary-access workflow, exact group or opening assignments, time-window configuration, hard-expiry fields, controller acknowledgements and local cache status. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the named person or approved cohort, precise openings and dependencies, one-off effective window, sponsor, purpose and all controllers that must enforce it. Timing/aggregation: at approval, before activation, at start, near expiry and after controller acknowledgement; show status by case and controller, not movement events; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate start/end boundary logic with synthetic cases, verify group-to-opening mapping, compare server and controller configuration hashes and align all clocks to the approved time source; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: offline controllers, nested groups, clock drift and global modes can widen or delay the intended window; technical expiry does not establish that a credential was returned.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the sponsor owns purpose and timing; area owners own destinations; Security owns temporary-access policy; PACS operations owns implementation and controller state. Current scope-matched approval with explicit dates and the acknowledged controller state outrank free-text requests, calendar invitations, prior exceptions and central status alone. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to the sponsor, area owner, Security or PACS operations; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, temporal comparison and recommendation. Decision interval: at approval, immediately before the window and at expiry review. Minimum evidence: verified identity or cohort, approved purpose, exact openings, start and end times with time zone, prerequisites, group graph and controller coverage. Method: intersect the approved scope with current route dependencies, calculate the effective window through nested paths and identify any excess or unbounded entitlement. Uncertainty: ambiguous time zone, inherited groups, clock offset, global modes, offline controllers and changes approved after the original case snapshot. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: do not recommend activation where any path lacks a bounded end or current acknowledgement; return HOLD and leave access unchanged.

ANSWER BOUNDARY

It must not activate, extend or expire the permission, infer attendance or treat an event booking as access approval. It must not approve a visitor, contractor, event or working arrangement; create or activate schedules; infer attendance or route; promise entry; or override Security, sponsor, area-owner or incident decisions.

WHY THIS IS USEFUL

The administrator can verify that a one-off case is technically bounded before an authorised colleague implements it.

Schedules, temporary access and visitors

AC-031 to AC-040

AC-033

QUESTION 3 OF 10

STAKEHOLDER QUESTION

What minimum visitor route, door set and credential validity window match the sponsor-approved visit?

SENSORS AND TELEMETRY

Proposed evidence and location: visitor-management case data, sponsor approval, approved route and escort status, temporary badge assignment, group/schedule configuration and controller acknowledgements; no camera or live tracking is used. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the visitor case, arrival and departure window, host or escort, approved entrance and destinations, route dependencies and the issued badge if any.

Timing/aggregation: at pre-registration, arrival processing, material itinerary change and expected departure; report case readiness and route coverage, not a visitor journey; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate visitor and PACS identifiers, test representative route groups with a non-person credential, verify clock and time-zone alignment and reconcile badge custody at issue and return; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a pre-registered route may be closed or changed, sponsor approval may lapse, and an access transaction does not prove escort presence or physical passage.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the sponsor owns visit purpose and itinerary; reception or Security owns visitor process and badge custody; area and route owners own destinations; Security owns visitor access policy. Current sponsor approval, visitor procedure, route-owner records and hard time limits outrank meeting invitations, prior visits, access events and the May 2021 floor plan. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: An approval or case record is stale when expired, superseded, outside its effective window, or missing a current owner/version; a connector state beyond the owner-approved freshness limit is UNKNOWN. Show conflicting values side by side and route them to the sponsor, reception, route, area or Security owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, graph reachability and recommendation. Decision interval: during pre-registration, at arrival and after a route or timing change. Minimum evidence: verified visitor case, accountable sponsor, approved destinations and route, escort rule, exact validity window, current closures and badge custody plan. Method: calculate the smallest group and window that covers the owner-approved route, identify missing dependencies and keep escorted and unescorted rules distinct. Uncertainty: late arrival, route closure, unverified escort status, shared entrances, time-zone errors, offline controllers and unsynchronised visitor/PACS records. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: hold issue or return UNKNOWN where current route, sponsor or timing evidence is incomplete; route to reception or Security.

ANSWER BOUNDARY

It must not identify or track the visitor, verify escort presence from events, decide the visit is legitimate, issue the badge or disclose the itinerary beyond authorised roles. It must not approve a visitor, contractor, event or working arrangement; create or activate schedules; infer attendance or route; promise entry; or override Security, sponsor, area-owner or incident decisions.

WHY THIS IS USEFUL

The administrator can prepare a visitor credential that reaches only the approved places for only the approved visit window.

AC-034

QUESTION 4 OF 10

STAKEHOLDER QUESTION

Does this contractor's temporary access align with the permit, work zone, working hours, escort condition and site rules?

SENSORS AND TELEMETRY

Proposed evidence and location: approved work order or permit status, contractor registry, induction and competency status, escort requirement, temporary group and schedule configuration and controller acknowledgement.

Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the named contractor or approved crew, work area and route, permit window, required plant or service zones, escort conditions and every affected opening. Timing/aggregation: at permit approval, before each work period when required, after scope change and at expiry; aggregate crew readiness by case while preserving individual details only for authorised processing; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate identifiers across contractor, permit and PACS systems, test date and shift boundaries, verify route mappings and controller clocks, and reconcile temporary credentials to custody records; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a valid permit does not prove the work area is currently safe or available; scope changes, permit suspensions and after-hours conditions may not propagate immediately.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the contractor manager owns crew identity and sponsorship; the permit issuer and competent area owner own work scope and conditions; Security owns access rules; reception or Security owns temporary credentials. The current permit and owner-approved work conditions outrank contractor schedules, purchase orders, prior access, observed presence and historic room labels. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A feed is stale after its owner-approved freshness limit; if no limit is documented, more than two missed scheduled updates, a schema change, or an unexplained count discontinuity makes the affected result UNKNOWN. Show conflicting values side by side and route them to the contractor manager, permit issuer, area owner or Security; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: comparison and rules-based validation. Decision interval: at access preparation, before each authorised work window where policy requires and after any permit change. Minimum evidence: verified contractor identity, current permit/work order, approved area and route, hours, escort status, prerequisites, credential and hard expiry. Method: compare every access object and time interval with the permit conditions, identify missing or excess scope and separate access readiness from permit validity. Uncertainty: feed delay, work-area changes, permit suspension, mixed crew roles, unverified route status, controller lag and local site rules outside PACS. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return NOT READY or CONFLICTED and keep access unchanged until the competent owner resolves the permit or route gap.

ANSWER BOUNDARY

It must not validate the work itself, certify competence or site safety, waive escort requirements, track the contractor or implement the access. It must not approve a visitor, contractor, event or working arrangement; create or activate schedules; infer attendance or route; promise entry; or override Security, sponsor, area-owner or incident decisions.

WHY THIS IS USEFUL

The administrator can prevent a contractor credential from exceeding the authorised task, location or shift.

Schedules, temporary access and visitors

AC-031 to AC-040

AC-035

QUESTION 5 OF 10

STAKEHOLDER QUESTION

Can this event be supported through temporary entrance permissions while excluding back-of-house and unrelated restricted areas?

SENSORS AND TELEMETRY

Proposed evidence and location: authoritative event booking and operating plan, approved public route, temporary entrance group, schedule and global-mode configuration, visitor or staff role assignments and controller acknowledgements. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the event entrance, public destination and egress route records, front-of-house support areas explicitly approved, excluded back-of-house zones and every opening touched by the temporary design. Timing/aggregation: during event planning, after route or timing change, before opening and after expiry; report group coverage and exception counts, not attendee movement; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate event and asset identifiers, test the temporary group with non-person credentials at representative openings, verify controller time and mode state and check hard expiry through nested paths; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: access configuration cannot establish crowd capacity, route safety, staffing, current closures or emergency suitability; shared groups may make clean separation impossible.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the event owner owns purpose and timing; building, Security and front-of-house owners own operating plan and route; Estates and competent safety owners own current closures and safety conditions. The approved current event and route plan outrank room bookings alone, previous events, access-event counts and the May 2021 floor plan. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: An approval or case record is stale when expired, superseded, outside its effective window, or missing a current owner/version; a connector state beyond the owner-approved freshness limit is UNKNOWN. Show conflicting values side by side and route them to the event, building, route, safety or Security owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: comparison, set calculation and recommendation. Decision interval: during event design, immediately before operation and at expiry review. Minimum evidence: approved event plan, exact entrances and public destinations, current route and closure records, staffing or escort rules, temporary group graph, schedule and expiry. Method: compare candidate groups with the approved public route, quantify back-of-house exposure, identify shared dependencies and recommend only a technically separable design. Uncertainty: temporary closures, shared entrances, global modes, unmapped internal doors, last-minute programme change and incomplete controller acknowledgement. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: do not recommend broad or inseparable access; return the design gap to event and Security owners and keep normal access unchanged.

ANSWER BOUNDARY

It must not certify event safety or capacity, choose public routes, infer crowd size, activate event mode or grant visitor access. It must not approve a visitor, contractor, event or working arrangement; create or activate schedules; infer attendance or route; promise entry; or override Security, sponsor, area-owner or incident decisions.

WHY THIS IS USEFUL

The administrator can support the event without accidentally extending temporary access into operational or restricted areas.

AC-036

QUESTION 6 OF 10

STAKEHOLDER QUESTION

Which active or future schedules require owner action for the upcoming closure, holiday or exceptional opening?

SENSORS AND TELEMETRY

Proposed evidence and location: current Estates closure calendar, Security operating-mode plan, PACS holiday and time-profile configuration, temporary entitlement register, future-effective changes and controller clock/acknowledgement state. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the affected building or zone, every schedule, exception, visitor or contractor case and controller whose effective interval overlaps the closure or exceptional opening. Timing/aggregation: when the authoritative calendar changes, at the agreed planning horizon, before the operating period and after final controller distribution; report impact by category and owner; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: test calendar and time-zone conversions, compare schedule versions, verify controller clocks and hashes and use synthetic boundary cases for overnight and holiday rules; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: calendar entries may be provisional, closures may have exceptions, emergency modes may override schedules and offline controllers may not receive the final plan.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Estates or the building owner owns closure/opening status; Security owns operating mode and exceptions; event, contractor and sponsor owners own their cases; PACS operations owns schedule implementation. The latest formally approved operating plan and scope-matched exception outrank general calendars, routine opening patterns, individual assumptions and historic building use. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to the building, Security, event, contractor or sponsor owner for each impacted item; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: forecast, comparison and recommendation. Decision interval: at planning updates, immediately before the period and after material change. Minimum evidence: approved closure or opening scope and dates, current schedule and exception inventory, owner assignments, controller coverage and emergency-mode constraints. Method: calculate interval overlap, classify permissions as unaffected, expected exception, owner review or conflict, and forecast unresolved work before the period starts. Uncertainty: provisional calendars, late exceptions, time-zone boundaries, controller lag, global modes and current operational status outside access control. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: make no schedule change; mark unresolved overlaps for owner decision and treat unacknowledged controllers as UNKNOWN.

ANSWER BOUNDARY

It must not declare a building closed or open, cancel access, approve exceptions, predict attendance or override emergency procedures. It must not approve a visitor, contractor, event or working arrangement; create or activate schedules; infer attendance or route; promise entry; or override Security, sponsor, area-owner or incident decisions.

WHY THIS IS USEFUL

The administrator can focus pre-closure checks on the schedules and exceptions that genuinely overlap the operating change.

Schedules, temporary access and visitors

AC-031 to AC-040

AC-037

QUESTION 7 OF 10

STAKEHOLDER QUESTION

Will clock drift, time-zone settings or a daylight-saving transition change any active or scheduled access window?

SENSORS AND TELEMETRY

Proposed evidence and location: NTP or approved time-source status, controller and server clock-offset samples, time-zone and daylight-saving configuration, schedule compiler output and timestamped test transactions. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: all controllers and integration services enforcing schedules in the affected time zone, plus every active or future profile spanning the tested boundary. Timing/aggregation: clock health at configured heartbeat, offset samples at the owner-defined interval and focused tests before daylight-saving or time-zone change; aggregate offsets by controller; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: compare against the approved time source, verify leap and daylight-saving rule version, use boundary test schedules and record measurement uncertainty and network delay; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: an apparently small offset can interact with overnight windows; disconnected controllers may use local clocks, vendor rules can differ and delayed events can mimic clock error.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: IT time-service owner owns the reference source; PACS operations owns controller time configuration; Security owns schedule policy; service vendors own supported time behaviour. The approved reference clock and tested controller behaviour outrank server display time, event order without offsets, device defaults and assumed daylight-saving support. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to the time-service owner, PACS operations, Security or relevant vendor; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: calculation, comparison and forecast. Decision interval: continuously for health and before any known time-rule boundary or schedule-critical event. Minimum evidence: approved time source, measured offset and uncertainty, controller time-zone/rule version, schedule set, connectivity state and controlled boundary test. Method: calculate effective start/end times under each controller rule, compare offsets with tolerance and forecast windows that may open early, close late or shift. Uncertainty: measurement latency, disconnected local clocks, vendor-specific daylight-saving handling, queued events and incomplete test coverage. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: treat affected schedules as UNKNOWN, leave them unchanged and route controllers outside tolerance for technical correction before reliance.

ANSWER BOUNDARY

It must not adjust clocks, alter schedules, infer who was affected from event times or certify the timing service. It must not approve a visitor, contractor, event or working arrangement; create or activate schedules; infer attendance or route; promise entry; or override Security, sponsor, area-owner or incident decisions.

WHY THIS IS USEFUL

The administrator can prevent one-hour or drift-related access errors before they appear as unexplained denials or grants.

AC-038

QUESTION 8 OF 10

STAKEHOLDER QUESTION

Is this temporary-access extension still supported by a current sponsor and purpose, and what is the shortest justified end time?

SENSORS AND TELEMETRY

Proposed evidence and location: current temporary entitlement, original and proposed end dates, sponsor and purpose workflow, prerequisite validity, exception history, group graph and controller distribution status. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the specific temporary case, all direct and inherited permissions, the sponsor and purpose evidence, current operating constraints and every controller enforcing the entitlement. Timing/aggregation: when an extension is requested, after sponsor or purpose change and immediately before the current expiry; show proposed interval and affected objects; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate effective dates and nested expiry with synthetic cases, verify sponsor authority and approval version, compare controller clocks and confirm no path lacks a hard end; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: operational delay alone is not authorisation, repeated extensions can conceal a continuing need, and offline controllers can retain a previous end state.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the sponsor owns continuing purpose; area and training owners own prerequisites; Security owns extension and exception policy; PACS operations owns implementation. A fresh scope-matched approval and current prerequisite evidence outrank the original request, prior extensions, system usage and a requester's preferred end date. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: An approval or case record is stale when expired, superseded, outside its effective window, or missing a current owner/version; a connector state beyond the owner-approved freshness limit is UNKNOWN. Show conflicting values side by side and route them to the sponsor, area, prerequisite or Security owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, temporal comparison and recommendation. Decision interval: on extension request and immediately before existing expiry. Minimum evidence: verified case and identity, current entitlement graph, original authority, fresh sponsor approval, continuing purpose, prerequisites and proposed hard end. Method: compare requested continuation with current evidence, calculate the shortest interval fully covered by authority and expose repeated or open-ended patterns for owner review. Uncertainty: late approvals, changed purpose, overlapping cases, nested groups, unrecorded work delay and controller propagation uncertainty. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: recommend no extension where authority or hard expiry is incomplete; allow the existing authorised expiry to stand.

ANSWER BOUNDARY

It must not decide the work merits an extension, renew access from observed use, contact the sponsor automatically or implement the new end time. It must not approve a visitor, contractor, event or working arrangement; create or activate schedules; infer attendance or route; promise entry; or override Security, sponsor, area-owner or incident decisions.

WHY THIS IS USEFUL

The administrator can route a tightly bounded extension instead of converting temporary access into an indefinite entitlement.

Schedules, temporary access and visitors

AC-031 to AC-040

AC-039

QUESTION 9 OF 10

STAKEHOLDER QUESTION

How should this multi-day permission represent different doors, hours or escort conditions on each day?

SENSORS AND TELEMETRY

Proposed evidence and location: structured day-by-day approval, time-profile components, group assignments, escort-condition flags, route mappings and controller acknowledgements; no live escort tracking is used. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: every approved day, interval, opening, route dependency and escort condition for the named case, including gaps between days and any overnight boundary. Timing/aggregation: during case design, after itinerary change and before each affected period; display an interval matrix and effective configuration by day; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: test interval boundaries, overnight splits, holiday interaction and nested group expiry with synthetic cases; verify time zones, controller clocks and configuration hashes; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: PACS may not natively enforce escort conditions, complex schedules can be misread, and a credential transaction cannot prove that an escort was present.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the sponsor owns purpose and itinerary; Security and area owners own access and escort rules; reception or contractor management owns operational escort process; PACS operations owns schedule representation. The explicit current day-level approval and operational escort procedure outrank a single broad date range, calendar shorthand and assumptions from prior visits. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to the sponsor, area, Security or operational escort owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: calculation, comparison and recommendation. Decision interval: when designing the case, after any day-level change and before implementation. Minimum evidence: approved day-by-day matrix, exact doors and dependencies, time zone, escort conditions, hard end dates, current group and schedule capabilities. Method: translate each approved interval into the smallest supported configuration, identify conditions PACS cannot enforce and keep those as explicit operational dependencies. Uncertainty: complex overlapping windows, overnight rules, unsupported escort logic, controller limits, daylight-saving change and ambiguous itinerary updates. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: do not flatten the case into a broader window; return unsupported conditions to the operational owner and leave access unchanged.

ANSWER BOUNDARY

It must not infer escort presence, certify the itinerary, create a broad all-days entitlement or claim that PACS enforces conditions it cannot observe. It must not approve a visitor, contractor, event or working arrangement; create or activate schedules; infer attendance or route; promise entry; or override Security, sponsor, area-owner or incident decisions.

WHY THIS IS USEFUL

The administrator can represent variable days accurately and make non-technical escort controls visible rather than silently losing them.

AC-040

QUESTION 10 OF 10

STAKEHOLDER QUESTION

Does every temporary entitlement have an enforceable hard expiry across direct, inherited and nested group paths?

SENSORS AND TELEMETRY

Proposed evidence and location: current entitlement and group graph, direct and inherited effective dates, exception register, local controller cache and distribution acknowledgements across all temporary access cases. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: every temporary credential or identity assignment in scope, all nested groups, schedules, controller-local rules and exceptions through which access could continue. Timing/aggregation: at creation, extension, periodic control and after expiry; report counts by bounded, conflicting, queued, offline and unbounded path; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate graph traversal with nested and cyclic test cases, verify date propagation and null handling, compare central and controller configurations and align expiry clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: some systems apply expiry only to the top-level assignment, local controllers may cache old data, and manual exceptions can sit outside the main graph.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Security owns temporary-access and exception policy; PACS operations owns group implementation; sponsors and area owners own continuing purpose; controller owners own local enforcement status. The earliest current authoritative end date and every acknowledged enforcement path outrank a top-level expiry display, expected inheritance and absence of recent use. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to Security, PACS operations, the sponsor or the owner of an unbounded local path; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: graph traversal and calculation. Decision interval: on every temporary lifecycle change and at the approved periodic control interval. Minimum evidence: complete group graph, assignment and exception dates, schedule rules, controller cache state, last acknowledgement and owner for every path. Method: enumerate every path from identity or credential to opening, calculate its effective end and classify paths as bounded, conflicting, unknown or technically unbounded. Uncertainty: hidden local rules, cycles, null dates, conflicting end semantics, offline controllers, incomplete inventory and exceptions outside workflow. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: treat any unverified path as UNKNOWN, do not claim expiry is effective and route the exact path for owner-controlled correction.

ANSWER BOUNDARY

It must not expire or revoke access, infer that an unused path is harmless, or conceal an unbounded nested entitlement behind a top-level date. It must not approve a visitor, contractor, event or working arrangement; create or activate schedules; infer attendance or route; promise entry; or override Security, sponsor, area-owner or incident decisions.

WHY THIS IS USEFUL

The administrator can find technical paths that would otherwise outlive the authorised temporary case.

Reader, controller and integration faults

AC-041 to AC-050

AC-041

QUESTION 1 OF 10

STAKEHOLDER QUESTION

Which readers, door interfaces or controllers are unhealthy or offline now, and which controlled openings are inside the impact boundary?

SENSORS AND TELEMETRY

Proposed evidence and location: reader and controller heartbeats, communications status, panel power and UPS health, network-port state, door-interface diagnostics, event-queue depth and current asset mapping. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: every commissioned reader, controller and door interface in the queried building or service boundary, including the openings and integrations dependent on each component. Timing/aggregation: at each configured heartbeat, normally tens of seconds to a few minutes, with status aggregated by asset and incident window; late events retain original and receive times; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate heartbeat logic against planned disconnect tests, verify network and power telemetry with competent measurements, reconcile asset labels and check clock offset and sequence continuity; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a missing heartbeat can reflect network, monitoring or maintenance state rather than device failure; health telemetry does not establish door security or physical usability.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: PACS operations owns device and service health; IT Network owns connectivity; Estates and approved maintainers own power and door hardware; Security owns operational response and impact tolerance. Current commissioned asset mapping and corroborated component health outrank a single monitoring alarm, help-desk description, central status icon and historical room location. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to PACS operations, IT Network, Estates, the maintainer or Security according to the failed layer; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: observation, dependency lookup and diagnosis. Decision interval: continuously for monitoring and at each incident triage update. Minimum evidence: asset inventory, expected heartbeat, last contact, network and power state, controller and queue status, maintenance window and mapped controlled openings. Method: correlate independent health layers, calculate the dependency impact boundary and rank monitoring, network, power, controller or interface fault candidates. Uncertainty: planned maintenance, intermittent connectivity, stale mapping, monitoring blind spots, shared controllers and uninstrumented lock hardware. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: report UNKNOWN where fewer than two independent evidence layers exist, avoid declaring an opening secure or failed and route to the responsible service.

ANSWER BOUNDARY

It must not unlock, lock or test doors remotely, infer who is affected from access records, certify physical security or substitute telemetry for an on-site safety check. It must not declare an unverified root cause, run intrusive tests outside approved procedure, change configuration, bypass controls, disclose person-level event histories or control an opening.

WHY THIS IS USEFUL

The administrator can triage the real asset footprint and direct field response without treating every missed heartbeat as the same fault.

AC-042

QUESTION 2 OF 10

STAKEHOLDER QUESTION

Why is this credential accepted at comparable doors but denied at this reader during the same authorised interval?

SENSORS AND TELEMETRY

Proposed evidence and location: authorised diagnostic transactions for the same token at the reported reader and an approved comparison reader, deny codes, entitlement and schedule snapshots, reader/controller configuration, clock status and local mode. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the affected token case, the reported reader and controller, one suitable comparison opening, all permission paths, schedule conditions and the short diagnostic window. Timing/aggregation: around the reported failure and during controlled retest; retain only the minimum event fields in the case and compare configuration versions at the same effective time; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: use a designated test credential where possible, verify reader and door mapping, confirm deny-code dictionary, compare controller hashes and measure clock offset against the approved source; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: acceptance elsewhere does not prove this opening should grant access; route groups, local modes, anti-passback state and hardware sensitivity can differ.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the sponsor or area owner owns eligibility; Security owns policy; PACS operations owns entitlement, schedule and controller configuration; approved maintainers own reader hardware. The approved current entitlement for this opening and controlled diagnostic evidence outrank anecdotal comparison, event absence and assumptions from another door. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to the area owner, PACS operations, Security or reader maintainer; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: diagnosis by comparison. Decision interval: when the fault is reported, after controlled retest and before routing the case. Minimum evidence: verified token and reader, current entitlement and schedule, deny code, controller mode and clock, comparison test and configuration versions. Method: compare entitlement path, schedule, local rule, token read and reader behaviour in a fixed sequence; return supported candidates and disconfirming evidence. Uncertainty: intermittent radio read, clock skew, anti-passback state, unrecorded local mode, stale controller copy and non-equivalent comparison opening. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: make no entitlement change; return UNKNOWN or the narrow supported service route when tests cannot isolate the layer.

ANSWER BOUNDARY

It must not infer attempted misuse, disclose the credential holder's other locations, reset anti-passback state or alter the reader or entitlement. It must not declare an unverified root cause, run intrusive tests outside approved procedure, change configuration, bypass controls, disclose person-level event histories or control an opening.

WHY THIS IS USEFUL

The administrator can distinguish a local technical denial from a genuine entitlement or schedule issue before escalating.

Reader, controller and integration faults

AC-041 to AC-050

AC-043

QUESTION 3 OF 10

STAKEHOLDER QUESTION

Do repeated forced-open or held-open alarms better support a hardware, contact-alignment, configuration or expected-use explanation?

SENSORS AND TELEMETRY

Proposed evidence and location: door-position contact, lock-monitor and request-to-exit states where fitted, configured alarm timers, controller events, maintenance tickets and owner-approved expected-use windows at the opening. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the named opening and its contact, lock, REX input, controller, alarm thresholds and maintenance history; only aggregate alarm counts and durations are used outside an authorised incident. Timing/aggregation: native state changes during the fault window, aggregated by five-minute or owner-approved intervals and compared with a recent healthy baseline after relevant works; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: witness open/closed contact alignment, validate REX and lock-monitor states, test alarm timers under approved procedure, verify door mapping and align controller clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: alarms do not reveal who opened a door or why; missing sensors, door bounce, propping for authorised operations, airflow and mechanical wear can produce similar patterns.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Security owns alarm response policy; Estates and approved door maintainers own hardware; PACS operations owns timer and input configuration; the area owner owns expected operating use. Witnessed maintenance tests and current owner-approved operating periods outrank alarm count alone, inferred behaviour and assumptions from access-event sequences. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to Security, PACS operations, Estates, the maintainer or area owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: observation, comparison and diagnosis. Decision interval: during an alarm cluster, after maintenance and at fault review. Minimum evidence: verified opening and sensor mapping, raw state sequence, alarm settings, maintenance status, expected-use window and at least one controlled functional test. Method: compare alarm timing with contact, lock and REX states and known operating windows; rank hardware, alignment, timer, configuration and operational explanations without asserting cause. Uncertainty: uninstrumented physical passage, door bounce, manual override, incomplete maintenance data, changed use pattern and clock or sequence gaps. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: report UNKNOWN when controlled evidence is absent; do not attribute the alarm to a person or suppress it through analytics.

ANSWER BOUNDARY

It must not identify a user, infer intent or misconduct, disable alarms, change thresholds or certify the door secure. It must not declare an unverified root cause, run intrusive tests outside approved procedure, change configuration, bypass controls, disclose person-level event histories or control an opening.

WHY THIS IS USEFUL

The administrator can send a precise evidence pack to maintenance, configuration or operational owners instead of relying on alarm volume.

AC-044

QUESTION 4 OF 10

STAKEHOLDER QUESTION

Is this intermittently connected controller losing events, delaying acknowledgements or enforcing stale local configuration?

SENSORS AND TELEMETRY

Proposed evidence and location: controller heartbeat, packet loss and latency, event sequence numbers, local queue depth and storage, server receive times, last acknowledged configuration hash, reboot and watchdog status. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the affected controller, all attached readers and openings, its network path, queued changes and the diagnostic interval before, during and after each disconnection. Timing/aggregation: heartbeat and queue metrics at the configured interval, raw sequence and send/receive times for the incident window, and daily gap summaries for recurrent faults; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate sequence continuity with controlled disconnect tests where approved, compare local and server hashes, verify network telemetry and align controller and collector clocks with measured offset; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: buffer behaviour varies by model and firmware; missing sequence numbers can reflect filtering or collector loss, and an acknowledgement may not prove correct physical enforcement.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: PACS operations owns controller and queue behaviour; IT Network owns connectivity; the vendor or maintainer owns firmware interpretation; Security owns operational response. Controller-local evidence, server receipt lineage and approved vendor behaviour outrank central status icons, user reports and assumptions from event arrival order. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to PACS operations, IT Network, the vendor or Security; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: observation, calculation and diagnosis. Decision interval: during each connectivity incident and after recovery reconciliation. Minimum evidence: controller ID and firmware, expected heartbeat, network evidence, sequence range, queue state, configuration hashes, reboot history and attached asset map. Method: separate transport loss, delayed buffer flush, collector loss and stale configuration by correlating sequence, queue, hash and network evidence; quantify unobserved intervals. Uncertainty: clock skew, undocumented filtering, queue rollover, partial local storage, firmware-specific acknowledgement and monitoring loss on the same network path. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: treat enforcement and event completeness as UNKNOWN for the gap, avoid backfilling assumptions and route for technical or on-site validation.

ANSWER BOUNDARY

It must not claim events never occurred, infer passage from buffered records, push configuration or reboot the controller. It must not declare an unverified root cause, run intrusive tests outside approved procedure, change configuration, bypass controls, disclose person-level event histories or control an opening.

WHY THIS IS USEFUL

The administrator can distinguish data delay from stale enforcement and prioritise the right recovery check.

Reader, controller and integration faults

AC-041 to AC-050

AC-045

QUESTION 5 OF 10

STAKEHOLDER QUESTION

Has identity-directory synchronisation degraded, and which queued requests or lifecycle actions may depend on stale identity data?

SENSORS AND TELEMETRY

Proposed evidence and location: connector job status, queue age, rejected-record logs, source and target record counts, checksums, schema versions and last successful watermark across identity, HR or Student and access systems. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the affected connector and data domain, all open access requests, credential lifecycle jobs and identities whose source change falls after the last trusted watermark. Timing/aggregation: at each scheduled sync, on error and during recovery; report aggregate impacted-case counts and authorised case identifiers, not unrelated identity attributes; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate field mappings and schema versions with controlled records, compare source/target counts and checksums, verify retry behaviour and align source, middleware and PACS clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: count agreement can hide field-level error, queue success can precede PACS processing, and not every source change is relevant to access eligibility.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the source identity owner owns authoritative records; IAM owns directory transformation; integration operations owns the connector; Security and credential teams own affected access decisions. The source-system watermark and verified transformation lineage outrank the directory copy, successful job status alone and manual assumptions that a delay has no material effect. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A feed is stale after its owner-approved freshness limit; if no limit is documented, more than two missed scheduled updates, a schema change, or an unexplained count discontinuity makes the affected result UNKNOWN. Show conflicting values side by side and route them to the source owner, IAM, integration operations or Security; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: observation, diagnosis and impact estimate. Decision interval: at every failed or delayed sync and before affected cases proceed. Minimum evidence: source watermark, last successful target commit, schema and mapping version, queue/rejection evidence, affected field classes and open-case timestamps. Method: identify the trustworthy cut-off, compare relevant source changes after it and estimate the affected case set with lower and upper bounds. Uncertainty: partial field failure, duplicate retries, unknown downstream cache, source corrections, schema drift and cases with manual evidence. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: mark impacted fields STALE, pause analytical readiness for affected cases and route recovery without altering source or PACS records.

ANSWER BOUNDARY

It must not alter identity data, decide which source value is correct, expose bulk identity attributes or continue a case by assuming unchanged status. It must not declare an unverified root cause, run intrusive tests outside approved procedure, change configuration, bypass controls, disclose person-level event histories or control an opening.

WHY THIS IS USEFUL

The administrator can isolate the requests that genuinely depend on the degraded feed instead of stopping or trusting everything blindly.

AC-046

QUESTION 6 OF 10

STAKEHOLDER QUESTION

How stale are the HR, student, visitor, contractor and training feeds used by current access decisions?

SENSORS AND TELEMETRY

Proposed evidence and location: source freshness watermarks, scheduled update times, connector health, schema and mapping versions, record-count continuity and last successful target commit for each authoritative feed. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: every configured authoritative feed used by access rules, the fields it supplies, open decisions depending on those fields and the downstream copies that receive them. Timing/aggregation: at each expected update and on demand; show freshness, last trusted interval and dependent-case counts by feed and field class; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate update cadence with source owners, test watermarks and failure alerts, compare source/target checksums and verify clock alignment across every hop; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: fresh transport does not guarantee correct content, feeds have different business cadences and a source may intentionally have no change during an interval.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: HR, Student Records, visitor, contractor and training owners own content; IAM and integration operations own transformations; Security owns how freshness affects access decisions. Owner-declared cadence and source evidence outrank a generic platform SLA, file modification time, dashboard green state and assumptions based on record counts alone. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A feed is stale after its owner-approved freshness limit; if no limit is documented, more than two missed scheduled updates, a schema change, or an unexplained count discontinuity makes the affected result UNKNOWN. Show conflicting values side by side and route them to the relevant source owner, IAM, integration operations or Security; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, comparison and calculation. Decision interval: continuously at expected updates and before a decision uses the field. Minimum evidence: documented cadence and freshness threshold, source and target watermarks, mapping version, last successful commit, quality alerts and dependent-case inventory. Method: calculate age and missed-update count for each field class, distinguish expected quiet periods from failed updates and map affected decisions. Uncertainty: unknown source cadence, silent schema change, partial field lag, clock skew, manual corrections and downstream caching. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: label the feed or field UNKNOWN where freshness cannot be established and prevent the analysis from presenting its dependent case as verified.

ANSWER BOUNDARY

It must not infer that unchanged data is correct, compare sensitive populations for surveillance, or override source-owner interpretation of freshness. It must not declare an unverified root cause, run intrusive tests outside approved procedure, change configuration, bypass controls, disclose person-level event histories or control an opening.

WHY THIS IS USEFUL

The administrator can see which decisions are evidence-safe and which must wait for a specific source recovery.

Reader, controller and integration faults

AC-041 to AC-050

AC-047

QUESTION 7 OF 10

STAKEHOLDER QUESTION

Can reader, controller and integration events be aligned to a trustworthy common timeline for this fault investigation?

SENSORS AND TELEMETRY

Proposed evidence and location: NTP or approved time-source status, measured clock offsets, controller event time, server receive time, integration processing time, sequence identifiers and log-pipeline latency. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the devices and services relevant to the authorised incident or fault window, including every clock domain and buffering step between event generation and retrieval. Timing/aggregation: offset health at the configured interval, with denser samples around the incident; preserve raw timestamps, receive times and uncertainty rather than rewriting them; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: measure offsets against the approved reference, validate time-zone and daylight-saving rules, test known event sequences and quantify network and buffering delay; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: some controllers timestamp only locally, old devices may lack reliable synchronisation, buffered events can arrive out of order and timestamp precision can differ.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: IT time-service owner owns reference time; PACS operations owns controller clocks; integration operations owns processing timestamps; Security or incident authority owns the authorised use. Measured offset and original timestamp provenance outrank display order, file order, server retrieval time and assumptions that all systems share one clock. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to the time-service, PACS, integration or incident owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: calculation and timeline reconstruction. Decision interval: during authorised technical diagnosis or incident evidence preservation. Minimum evidence: asset IDs, clock source and offset measurements, raw event/receive/process times, sequence IDs, buffer state and log provenance for the defined window. Method: convert to a common reference while retaining original values, propagate offset and latency uncertainty, and order only events whose intervals do not overlap ambiguously. Uncertainty: unmeasured drift, coarse timestamp precision, queueing, clock resets, daylight-saving errors, log filtering and missing sequence data. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: present an interval order or UNORDERED result rather than a false exact sequence; preserve originals and route critical ambiguity to the incident owner.

ANSWER BOUNDARY

It must not infer a person's route, intent or physical passage from event order, or alter source logs to create a cleaner timeline. It must not declare an unverified root cause, run intrusive tests outside approved procedure, change configuration, bypass controls, disclose person-level event histories or control an opening.

WHY THIS IS USEFUL

The administrator can explain which technical events are reliably ordered and which remain temporally ambiguous.

AC-048

QUESTION 8 OF 10

STAKEHOLDER QUESTION

After the outage, which controllers recovered fully, which events arrived late, and which local permissions remain unverified?

SENSORS AND TELEMETRY

Proposed evidence and location: controller reboot and heartbeat, queue and buffer flush status, event sequence ranges, configuration hashes, distribution acknowledgements, network recovery, approved test transactions and door-state checks where fitted. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: every controller affected by the outage, attached readers and openings, queued access changes, local configurations and the event interval from last trusted contact through recovery. Timing/aggregation: continuously through recovery, with a formal reconciliation snapshot after queues settle and controlled tests complete; separate event completeness from permission enforcement; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: verify sequence ranges and buffer limits, compare pre/outage/post configuration hashes, use designated test credentials, witness door contacts where fitted and align clocks including reboot offset; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: full connectivity does not prove buffered completeness or current local enforcement; queues can overflow, events can duplicate and some doors lack monitored state.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: PACS operations owns controller recovery; IT Network owns connectivity; Security owns operational acceptance; approved maintainers own hardware tests; authoritative request owners own queued changes. Reconciled local/controller evidence and controlled tests outrank a central green status, successful ping, late event volume and assumptions that all queued changes applied. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to PACS operations, IT Network, Security or the relevant maintainer; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: observation, reconciliation and comparison. Decision interval: during recovery and at formal return-to-service review. Minimum evidence: affected controller list, last trusted state, queue and sequence evidence, pre/post hashes, queued change IDs, network state and controlled test results. Method: classify each controller separately for connectivity, event completeness, configuration currency and functional test; calculate known gap intervals and duplicate ranges. Uncertainty: queue rollover, unmeasured reboot clock shift, incomplete test coverage, local manual changes, missing door contacts and delayed source updates during outage. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: keep the controller or affected functions UNVERIFIED, avoid claiming full recovery and route remaining checks before normal reliance.

ANSWER BOUNDARY

It must not backfill missing events, infer passage, push queued changes, declare an opening secure or return a system to service. It must not declare an unverified root cause, run intrusive tests outside approved procedure, change configuration, bypass controls, disclose person-level event histories or control an opening.

WHY THIS IS USEFUL

The administrator can distinguish apparent connectivity recovery from evidence that configuration and event handling are actually reconciled.

Reader, controller and integration faults

AC-041 to AC-050

AC-049

QUESTION 9 OF 10

STAKEHOLDER QUESTION

Are these denials consistent with anti-passback or area-state logic whose recorded state is incomplete or inconsistent?

SENSORS AND TELEMETRY

Proposed evidence and location: anti-passback state-machine configuration, reader direction mapping, controller-local state, reset and override logs, door contact and REX states where fitted, and controlled non-person test sequences. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the defined anti-passback area, every entrance and exit reader, controller state, reset authority and the short denial window; person identities are masked unless the authorised case requires one token. Timing/aggregation: on fault report and during controlled testing; aggregate denial reasons and state conflicts by area and interval, not individual movement histories; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate reader direction and area mapping, verify door and REX inputs, use designated test credentials, test reset logic under approval and align controller clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: system state does not prove physical presence; tailgating, manual exits, missing sensors, door propping and reader bypass can make logical and physical states diverge.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Security owns anti-passback policy and reset authority; PACS operations owns logic and state; Estates owns door hardware; the area owner owns operational constraints. Current commissioned mapping and controlled state-machine evidence outrank inferred routes, a single denial code and assumptions that recorded entry equals physical presence. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to Security, PACS operations, Estates or the area owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: diagnosis by state-machine comparison. Decision interval: when a repeated denial pattern is reported and after relevant configuration or hardware change. Minimum evidence: area and reader map, current state-machine configuration, controller state, deny codes, reset/override history, sensor mapping and controlled test result. Method: compare expected and observed logical transitions, identify impossible or missing transitions and rank mapping, state, hardware and user-process explanations without attributing behaviour. Uncertainty: unobserved passage, manual egress, tailgating, missing REX, clock drift, stale state and incomplete test sequence. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return UNKNOWN and route an authorised reset or mapping check; make no automatic state correction.

ANSWER BOUNDARY

It must not infer occupancy, accuse a person of tailgating, reset area state, disclose movement history or weaken anti-passback rules. It must not declare an unverified root cause, run intrusive tests outside approved procedure, change configuration, bypass controls, disclose person-level event histories or control an opening.

WHY THIS IS USEFUL

The administrator can identify a logical-state fault without treating access events as ground truth about where people are.

AC-050

QUESTION 10 OF 10

STAKEHOLDER QUESTION

Did the latest firmware or configuration deployment pass approved tests at every affected reader, interface and controller?

SENSORS AND TELEMETRY

Proposed evidence and location: signed deployment package and hash, rollout log, device firmware and configuration version, controller acknowledgement, automated health checks and controlled credential, door-contact, lock-monitor and REX tests where applicable. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: every reader, controller and controlled opening in the approved change scope, including canary assets, rollback group and integrations affected by the release. Timing/aggregation: before deployment, at each rollout stage, immediately after change and through the approved observation window; report pass, fail, not tested and not reachable by asset; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: verify package signature and hash, baseline and post-change versions, use designated test credentials, witness physical inputs and compare clocks and event sequence continuity; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: automated health does not exercise every physical path; a test credential may not represent all technologies, and unchanged faults can be wrongly attributed to the release.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the change authority and Security own approval and acceptance criteria; PACS operations owns deployment; maintainers and vendors own device tests; Estates owns physical interfaces. The approved change plan, signed package, asset-level version evidence and witnessed tests outrank rollout success percentage, vendor assurance and absence of help-desk calls. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to the change authority, Security, PACS operations, maintainer or Estates; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: comparison and recommendation. Decision interval: at each deployment gate and at the end of the observation window. Minimum evidence: approved scope and acceptance criteria, baseline versions, signed package, per-asset result, controlled test evidence, health window and rollback plan. Method: compare every affected asset with the planned version and required tests, separate untested from failed and recommend accept, hold or rollback review without executing it. Uncertainty: offline assets, partial test coverage, pre-existing defects, hidden dependencies, delayed failures and configuration not reflected in central inventory. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: mark the deployment PARTIALLY VERIFIED or FAILED where critical evidence is absent; stop the analytical recommendation at the approved gate.

ANSWER BOUNDARY

It must not deploy or roll back software, certify the release secure or safe, suppress failed tests or treat a successful central push as physical acceptance. It must not declare an unverified root cause, run intrusive tests outside approved procedure, change configuration, bypass controls, disclose person-level event histories or control an opening.

WHY THIS IS USEFUL

The administrator can present an asset-level acceptance pack and prevent untested openings from disappearing inside a high overall success rate.

Reviews, audit trails and least privilege

AC-051 to AC-060

AC-051

QUESTION 1 OF 10

STAKEHOLDER QUESTION

Which identities, templates, groups, zones and exceptions are due for review, and who is accountable for each decision?

SENSORS AND TELEMETRY

Proposed evidence and location: review-register status, owner and attestation assignments, identity and affiliation freshness, role/group/zone inventory, exception dates and configuration versions; no movement sensing is required. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the complete review population defined by policy, including every in-scope identity, role template, permission group, controlled zone, privileged account and exception with its accountable owner. Timing/aggregation: at the policy-defined review cycle, on owner or scope change and before closure; show due, overdue, blocked and complete counts by owner and object type; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate population extraction against source inventories, test owner-resolution rules, reconcile object counts and capture a signed snapshot with aligned retrieval times; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: an object can exist outside the main inventory, ownership may be shared or vacant, and recent modification time does not prove substantive review.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Security and governance owners define review scope and cadence; HR, Student Records and IAM own identity status; area owners own need; PACS operations owns configuration inventory. The current approved review policy and authoritative object inventories outrank previous review lists, email reminders, recent use and assumed ownership from naming conventions. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to the governance, identity, area, Security or PACS owner responsible for the object; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup and calculation. Decision interval: at review launch, during progress monitoring and at closure. Minimum evidence: current review policy and cut-off, complete source inventories, object owner, effective status, last substantive attestation and open exception or conflict. Method: construct the in-scope population at a fixed cut-off, assign each item to an accountable decision owner and calculate progress without interpreting the decision. Uncertainty: orphaned objects, duplicate inventories, owner absence, late source updates, disputed scope and attestations that lack evidence or correct authority. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: keep the item OPEN or BLOCKED, do not infer approval from silence and route ownership gaps to governance.

ANSWER BOUNDARY

It must not decide that an entitlement remains necessary, close a review from system activity, contact owners outside approved process or modify reviewed objects. It must not decide entitlement necessity from observed use, make HR, sponsor or Security decisions, expose movement histories or privileged audit data beyond authorised roles, or execute review actions.

WHY THIS IS USEFUL

The administrator can launch a complete review and direct each decision to an accountable owner instead of circulating an incomplete spreadsheet.

AC-052

QUESTION 2 OF 10

STAKEHOLDER QUESTION

Which active entitlements conflict with current role, sponsor, training, area-owner or end-date evidence?

SENSORS AND TELEMETRY

Proposed evidence and location: current entitlement snapshot, direct and inherited group graph, authoritative role and sponsor records, prerequisite status, area approvals, exception register and controller versions. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: every in-scope identity and all entitlements, including nested, temporary, lift and schedule paths, mapped to the evidence that currently supports or conflicts with each path. Timing/aggregation: at review cut-off and after a material source change; show counts and entitlement-level reasons, with identity detail only to authorised reviewers; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate mapping between source roles and access templates, test nested graph traversal, verify snapshot hashes and align effective and retrieval times across systems; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a mismatch can be legitimate through an approved exception, source lag or concurrent role; absence of use is not evidence that permission is unnecessary.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: HR, Student Records and sponsors own current role or purpose; training and area owners own prerequisites and destination need; Security owns role matrix and exception policy. Current source-owner evidence and approved exceptions outrank entitlement age, past usage, peer comparison and the permission currently present in PACS. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A feed is stale after its owner-approved freshness limit; if no limit is documented, more than two missed scheduled updates, a schema change, or an unexplained count discontinuity makes the affected result UNKNOWN. Show conflicting values side by side and route them to the role, sponsor, prerequisite, area or Security owner linked to the conflict; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: comparison. Decision interval: at each formal review cut-off and before any proposed remediation. Minimum evidence: verified identity, current affiliation and role, sponsor, prerequisites, area approvals, complete entitlement graph, exception authority and effective dates. Method: attribute every entitlement path to current evidence, classify supported, exception-supported, conflicting or unknown and preserve direct versus inherited source. Uncertainty: overlapping roles, source delay, unrecorded exceptions, ambiguous templates, local controller state and missing owner decisions. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: present candidates for owner review only, leave access unchanged and mark unsupported evidence CONFLICTED or UNKNOWN.

ANSWER BOUNDARY

It must not remove access because of a comparison result, infer necessity from activity, decide which source conflict is correct or disclose unrelated entitlements. It must not decide entitlement necessity from observed use, make HR, sponsor or Security decisions, expose movement histories or privileged audit data beyond authorised roles, or execute review actions.

WHY THIS IS USEFUL

The administrator can focus reviewers on evidence conflicts rather than asking them to inspect every entitlement without context.

Reviews, audit trails and least privilege

AC-051 to AC-060

AC-053

QUESTION 3 OF 10

STAKEHOLDER QUESTION

Which accounts can administer the PACS, what exact scope can each change, and when was that privilege last approved?

SENSORS AND TELEMETRY

Proposed evidence and location: PACS administrator directory, role-based administration scopes, MFA and account status, approval and recertification records, privileged session metadata and immutable administrative audit logs; no secret values are collected. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: all human and service accounts with configuration, identity, credential, reporting, integration or audit privileges, including delegated scopes and emergency accounts. Timing/aggregation: on account or role change, at the approved recertification interval and after a privileged incident; report scope and approval age, not session content; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate account inventory against platform and identity sources, test role-to-capability mapping with non-production accounts, verify log completeness and align identity, PACS and audit clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: an account may be technically active but operationally dormant or tightly controlled; MFA status does not prove authorised use, and vendor or service accounts may sit outside standard directories.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Security and Information Security own privileged-access policy; PACS service owner owns platform roles; IAM owns account status; account sponsors own service accounts; auditors own independent review requirements. Current approved privilege assignment and platform-enforced scope outrank job title, group name, past administrative activity and informal emergency practice. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to Security, Information Security, the PACS service owner or service-account sponsor; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, comparison and separation-of-duties review. Decision interval: on privileged change, at recertification and after suspected compromise. Minimum evidence: complete account inventory, account type and owner, enforced scope, approval, MFA/control state, last recertification, emergency conditions and audit coverage. Method: map each account to actual capabilities, compare with current approval and separation rules, and identify excessive, unowned, stale or unlogged privilege for owner decision. Uncertainty: hidden local accounts, shared emergency procedures, vendor access, role-name ambiguity, incomplete logging and temporary elevation not yet reconciled. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: do not disable accounts; mark unverified privilege HIGH-PRIORITY UNKNOWN and route for independent owner review.

ANSWER BOUNDARY

It must not reveal credentials or session contents, handle secrets, infer misuse from access time, approve privilege or change administrator roles. It must not decide entitlement necessity from observed use, make HR, sponsor or Security decisions, expose movement histories or privileged audit data beyond authorised roles, or execute review actions.

WHY THIS IS USEFUL

The administrator can recertify the real administrative capability surface rather than relying on account names or nominal roles.

AC-054

QUESTION 4 OF 10

STAKEHOLDER QUESTION

Which temporary, break-glass, project or emergency exceptions remain active beyond their approved end?

SENSORS AND TELEMETRY

Proposed evidence and location: exception register, effective start and end dates, direct and nested entitlement graph, approval and closure workflow, controller cache and acknowledgement state. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: every exception class in review scope, its owner, identity or group assignment, affected openings and all technical paths through which it can remain effective. Timing/aggregation: at exception creation, near expiry, after end date and at periodic review; aggregate overdue duration and path count by owner and risk class; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate hard-expiry propagation with nested test cases, reconcile exception IDs across workflow and PACS, compare controller hashes and verify clock/time-zone handling; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: an expired approval may still be technically active through inheritance or offline cache, while an active approval may have a legitimate incident extension not yet synchronised.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Security owns exception policy; incident authority owns emergency continuation; sponsors and project owners own purpose; PACS operations owns technical enforcement. Current explicit continuation authority and effective dates outrank the exception's original purpose, recent use, technical presence and verbal request. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to Security, incident authority, sponsor or PACS operations; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: calculation, graph traversal and comparison. Decision interval: daily or at the owner-defined control interval and immediately after an exception end date. Minimum evidence: exception ID and type, accountable owner, current authority, exact effective dates, complete entitlement paths, controller state and any approved extension. Method: calculate overdue time, enumerate active technical paths and distinguish authorised extension, queued expiry, offline uncertainty and unsupported continuation. Uncertainty: late incident records, nested groups, null dates, clock drift, offline controllers, manual local changes and incomplete exception inventory. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: leave the technical state unchanged, flag unsupported continuation for urgent owner action and treat offline enforcement as UNKNOWN.

ANSWER BOUNDARY

It must not close, revoke or extend an exception, infer that recent use validates it, or weaken emergency procedure. It must not decide entitlement necessity from observed use, make HR, sponsor or Security decisions, expose movement histories or privileged audit data beyond authorised roles, or execute review actions.

WHY THIS IS USEFUL

The administrator can expose exceptions that have outlived authority even when they are hidden through nested groups or local caches.

Reviews, audit trails and least privilege

AC-051 to AC-060

AC-055

QUESTION 5 OF 10

STAKEHOLDER QUESTION

Which active credentials are tied to ended affiliations, unresolved identity records or no accountable sponsor?

SENSORS AND TELEMETRY

Proposed evidence and location: credential-to-identity binding, authoritative affiliation and sponsor status, identity-resolution flags, token lifecycle state, entitlement snapshot and controller acknowledgement. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: all active token types in review scope, their verified identity, current affiliation or sponsored purpose, accountable owner, groups and controller enforcement coverage. Timing/aggregation: on affiliation end, identity conflict, sponsor change and at scheduled review; report counts by evidence gap and authorised case detail; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate identity linkage across platforms, reconcile token inventory, test end-date and sponsor propagation and verify source and controller clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: ended source status can be delayed or corrected, a token may have another valid affiliation, and active central status does not prove every controller accepts it.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: HR, Student Records or collaborator registry owns affiliation; IAM owns identity resolution; sponsors own purpose; credential office owns token status; Security owns action policy. Current authoritative identity, affiliation and sponsor evidence outrank credential activity, token age, directory display and assumptions from access events. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A feed is stale after its owner-approved freshness limit; if no limit is documented, more than two missed scheduled updates, a schema change, or an unexplained count discontinuity makes the affected result UNKNOWN. Show conflicting values side by side and route them to the affiliation, IAM, sponsor, credential-office or Security owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup and comparison. Decision interval: on lifecycle events and at the approved periodic review. Minimum evidence: stable token and identity IDs, complete current affiliations, sponsor, lifecycle state, entitlement paths, controller coverage and any continuing exception. Method: classify each token as supported, multi-affiliation supported, identity conflict, sponsor missing, affiliation ended or unknown without selecting an action. Uncertainty: source lag, duplicate identities, concurrent roles, unrecorded sponsor transfer, temporary grace periods and offline controller state. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: present the item for authoritative owner decision, make no revocation and mark unresolved identity linkage CONFLICTED.

ANSWER BOUNDARY

It must not decide employment or student status, infer token use or holder location, disable a credential or disclose the review outside authorised roles. It must not decide entitlement necessity from observed use, make HR, sponsor or Security decisions, expose movement histories or privileged audit data beyond authorised roles, or execute review actions.

WHY THIS IS USEFUL

The administrator can direct lifecycle review to credentials that have lost a clear authoritative basis without assuming every ended affiliation means immediate revocation.

AC-056

QUESTION 6 OF 10

STAKEHOLDER QUESTION

Which permission groups or zones are orphaned because no current owner, purpose or mapped opening can be evidenced?

SENSORS AND TELEMETRY

Proposed evidence and location: PACS configuration inventory, group and zone owner register, mapped door and lift assets, assignment counts, change history and configuration references from workflows and integrations.

Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: every active and future-effective group or zone, its nested dependencies, assigned identities or roles, controlled assets, schedules, integrations and declared purpose.

Timing/aggregation: at configuration change and periodic governance review; report orphan indicators and dependency counts, with person-level assignments available only to authorised reviewers; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates.

Quality, calibration and provenance: validate inventory completeness against controller and server configuration, verify asset mappings physically where needed, test nested graph extraction and compare owner records by version; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage.

Limits: a group with no assignments can still be a required template component; an assigned group may be dormant but legitimate, and naming does not establish purpose or ownership.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Security owns permission design and policy; PACS operations owns configuration; area, lift or building owners own operational purpose; Estates owns physical asset mapping; governance owns ownership requirements. Explicit current owner and purpose records plus commissioned asset mapping outrank group name, assignment count, last-use evidence and the May 2021 floor plan. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to Security, governance, PACS operations or the current asset/area owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, graph analysis and completeness comparison. Decision interval: at periodic review and before reuse, edit or retirement of the object. Minimum evidence: complete group/zone inventory, owner and purpose fields, mapped assets, dependencies, assignments, change history and current policy requirements. Method: classify missing owner, missing purpose, unmapped asset, unused reference and conflicting ownership separately; calculate the impact of any proposed quarantine or retirement. Uncertainty: hidden controller-local objects, legacy integrations, ambiguous names, unrecorded owners, temporary projects and incomplete physical mapping. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: mark the object ORPHAN-CANDIDATE only, prevent reuse by recommendation and route ownership or mapping validation before any change.

ANSWER BOUNDARY

It must not delete, disable or repurpose a group, infer purpose from usage or naming, or expose assigned identities outside the approved review. It must not decide entitlement necessity from observed use, make HR, sponsor or Security decisions, expose movement histories or privileged audit data beyond authorised roles, or execute review actions.

WHY THIS IS USEFUL

The administrator can prevent unowned configuration objects from being reused casually while preserving dependencies that still need investigation.

Reviews, audit trails and least privilege

AC-051 to AC-060

AC-057

QUESTION 7 OF 10

STAKEHOLDER QUESTION

Which entitlement combinations invoke separation-of-duties, dual-approval or mutually exclusive access rules?

SENSORS AND TELEMETRY

Proposed evidence and location: current entitlement graph, privileged role and conflict-rule library, approval workflow, account and identity linkage and exception records; no movement or productivity data is required. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the reviewed identity or service account, all direct and inherited permissions and every current separation or dual-approval rule applicable to the scope. Timing/aggregation: on request, role change, privileged recertification and policy update; show rule hits and evidence, not behavioural interpretation; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate rule versions and mappings with approved positive and negative test cases, verify identity/account linkage and compare effective dates across policy and entitlement snapshots; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a rule hit is not proof of wrongdoing, and business processes may include documented compensating controls or exceptions that are outside PACS.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Security and Information Security own conflict rules; business and area owners own process roles; governance owns compensating-control acceptance; IAM and PACS owners own identity and entitlement data. Current approved rule and explicit exception or compensating-control decision outrank role titles, observed activity, prior acceptance and system naming. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to Security, Information Security, governance or the relevant business owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: comparison and rules-based recommendation. Decision interval: at each relevant access change and formal privileged review. Minimum evidence: verified identity or account, complete entitlement graph, current conflict-rule version, required approvers and any current exception or compensating control. Method: evaluate each documented rule against direct and inherited permissions, show the exact combination and route to required independent approval without judging behaviour. Uncertainty: incomplete rule library, hidden external-system privilege, role ambiguity, stale exception and nested group paths missing from the snapshot. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return CONFLICTED or UNKNOWN, leave access unchanged and require the named independent decision owners.

ANSWER BOUNDARY

It must not accuse a person, infer fraud or intent, approve a conflict, inspect work content or alter entitlements. It must not decide entitlement necessity from observed use, make HR, sponsor or Security decisions, expose movement histories or privileged audit data beyond authorised roles, or execute review actions.

WHY THIS IS USEFUL

The administrator can route sensitive combinations through the right independent approval before they become hidden privilege conflicts.

AC-058

QUESTION 8 OF 10

STAKEHOLDER QUESTION

Is the administrative change chain complete from request and approval through implementation, acknowledgement and rollback evidence?

SENSORS AND TELEMETRY

Proposed evidence and location: workflow IDs, approval records, privileged administrative commands, configuration diffs, controller acknowledgements, deployment logs, rollback records and tamper-evident audit-log integrity checks. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the single change or approved batch, every affected object and controller, all human or service accounts that performed authorised steps and the defined pre/post evidence window. Timing/aggregation: at each change stage and formal closure; retain event-level detail under audit permission and publish only completeness status and counts outside the case; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: verify identifier propagation across systems, compare cryptographic log or file integrity where available, test clock alignment and sequence continuity and reconcile expected versus observed stage counts; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a logged command does not prove intended physical effect, acknowledgement semantics differ by platform and some manual field steps may require separate signed evidence.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the change owner owns scope and approval; Security and PACS operations own implementation; controller and integration owners own acknowledgement; auditors and Records Management own evidence requirements. Source-native signed or accountable records with linked IDs and times outrank copied screenshots, narrative recollection, log order without clock correction and a final-state snapshot alone. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to the change owner, Security, PACS operations, system owner, auditor or Records Management; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: traceability lookup and chain-of-custody comparison. Decision interval: during change execution and before administrative closure or audit release. Minimum evidence: approved change ID, scope and versions, accountable approvals, executor identity, command or diff, controller result, test evidence, rollback plan and log provenance. Method: build an ordered evidence chain, verify that every planned object and stage has a linked record and classify missing, conflicting, late or unverifiable evidence. Uncertainty: clock skew, manual steps, incomplete log retention, service-account indirection, vendor acknowledgement ambiguity and out-of-band emergency changes. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: keep the change record OPEN or EVIDENCE INCOMPLETE and preserve sources; do not invent or reconstruct missing approvals.

ANSWER BOUNDARY

It must not certify that the change was correct, infer intent from admin actions, expose privileged logs beyond authorised audit roles or execute rollback. It must not decide entitlement necessity from observed use, make HR, sponsor or Security decisions, expose movement histories or privileged audit data beyond authorised roles, or execute review actions.

WHY THIS IS USEFUL

The administrator can close only changes with a reproducible evidence chain and direct missing stages to the accountable owner.

Reviews, audit trails and least privilege

AC-051 to AC-060

AC-059

QUESTION 9 OF 10

STAKEHOLDER QUESTION

For this role change, does the proposed package remove obsolete access as reliably as it adds justified access?

SENSORS AND TELEMETRY

Proposed evidence and location: authoritative role-change record, before-and-proposed entitlement snapshots, direct and inherited group graph, exception dates, change workflow and controller acknowledgement plan. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the affected identity, old and new role evidence, every current entitlement path, every proposed path and the controllers enforcing them. Timing/aggregation: at change design, immediately before implementation and after propagation evidence; show add and remove sides with equal completeness measures; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate snapshot and graph completeness, test delta logic with overlapping-role cases, compare versions and align effective and controller times; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: some old permissions may remain valid through another role, removal can break route dependencies, and local or offline controller state may differ from the central plan.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: HR or Student Records owns role change; old and new managers own continuing duties; area owners own destination need; Security owns least-privilege policy; PACS operations owns implementation. Current role and owner evidence for each entitlement outrank blanket removal, copied old packages, observed non-use and the assumption that every old group is obsolete. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: An approval or case record is stale when expired, superseded, outside its effective window, or missing a current owner/version; a connector state beyond the owner-approved freshness limit is UNKNOWN. Show conflicting values side by side and route them to the old/new duty, area, Security or PACS owner linked to the disputed path; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: bidirectional comparison and recommendation. Decision interval: during role-change design, at approval and after authorised implementation. Minimum evidence: verified old and new roles, complete current entitlement graph, continuing exceptions, route dependencies, effective time and controller coverage. Method: attribute every current path to a continuing or ended basis, calculate add and remove candidates symmetrically and identify dependencies or conflicts before any action. Uncertainty: concurrent roles, undocumented continuing duties, nested groups, stale snapshots, route coupling and incomplete controller propagation. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: leave disputed entitlements unchanged, mark them OWNER REVIEW and make no change through the analytical service.

ANSWER BOUNDARY

It must not remove access because it is old or unused, decide that a duty ended, or trade a removal against an unrelated addition. It must not decide entitlement necessity from observed use, make HR, sponsor or Security decisions, expose movement histories or privileged audit data beyond authorised roles, or execute review actions.

WHY THIS IS USEFUL

The administrator can prevent role changes from becoming one-way privilege accumulation while protecting legitimate continuing access.

AC-060

QUESTION 10 OF 10

STAKEHOLDER QUESTION

Can I prepare a least-privilege evidence pack for this authorised review without exposing detailed movement records?

SENSORS AND TELEMETRY

Proposed evidence and location: current identity, role and entitlement snapshots, group and dependency graph, approvals, exceptions, owner attestations and aggregate event-quality indicators only where the lawful review purpose requires them. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the authorised review population and cut-off, current access objects and supporting authority; person-level door-event sequences are excluded unless separately authorised for a specific purpose. Timing/aggregation: at the fixed review cut-off and on evidence refresh; present entitlement and authority state, aggregate data-quality and age, and minimal redacted case detail; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate population and graph completeness, verify source versions and clocked cut-off, test redaction and role-based access, and record every transformation and export; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: lack of recent events is not evidence that access is unnecessary; aggregate activity can be incomplete because of offline devices or alternative routes.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the review owner and Security define purpose and scope; HR, Student Records, sponsors and area owners own supporting authority; Data Protection and Records Management own minimisation and retention. Current owner evidence and approved least-privilege rules outrank movement patterns, peer comparison, anecdotal use and data collected for another purpose. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to the review owner, Security, Data Protection, Records Management or relevant authority owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: comparison, redaction and recommendation. Decision interval: at review preparation, controlled refresh and final authorised export. Minimum evidence: documented purpose and recipients, fixed population and cut-off, current entitlement graph, authority evidence, exception records, provenance and approved redaction rules. Method: build an entitlement-to-authority matrix, minimise fields, replace identity detail with authorised identifiers where possible and include uncertainty and source-quality notes. Uncertainty: incomplete mappings, unresolved authority, small aggregates that risk re-identification, stale evidence and missing controller coverage. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: withhold or further aggregate unsafe fields, mark evidence gaps UNKNOWN and require Data Protection or review-owner decision before release.

ANSWER BOUNDARY

It must not use detailed movement history as a proxy for need, disclose identifiable event sequences, decide entitlements or certify least privilege. It must not decide entitlement necessity from observed use, make HR, sponsor or Security decisions, expose movement histories or privileged audit data beyond authorised roles, or execute review actions.

WHY THIS IS USEFUL

The administrator can give reviewers a useful evidence pack centred on authority and configuration rather than surveillance.

Incidents, suspension, revocation and recovery

AC-061 to AC-070

AC-061

QUESTION 1 OF 10

STAKEHOLDER QUESTION

For this lost, stolen or suspected-compromised credential, where is suspension confirmed and where is enforcement still uncertain?

SENSORS AND TELEMETRY

Proposed evidence and location: authorised incident instruction, central credential status, distribution queue, controller acknowledgements, offline-panel last contact, local cache version and replacement case status. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the affected physical or mobile token, all linked credentials only when the incident authority includes them, every controller enforcing its groups and the full instruction path. Timing/aggregation: from instruction time through each acknowledgement and at incident update intervals; report confirmed, queued, offline and unknown coverage without showing movement events; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate token and instruction identifiers, compare central and local status versions, verify queue sequence and clock offsets and test known suspension propagation in a controlled environment; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: central suspension does not prove local enforcement at an offline controller, and missing subsequent events does not prove containment or non-use.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Security or incident authority owns the containment decision; the credential office owns token lifecycle; PACS operations owns distribution; mobile-service and controller owners own platform enforcement. The verified incident instruction and component acknowledgement outrank user report alone, central display state, expected propagation and absence of access events. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to the incident authority, credential office, PACS operations or platform owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, observation and propagation estimate. Decision interval: immediately after instruction and at every incident status update. Minimum evidence: verified token ID, authoritative instruction and scope, central status, complete target component list, acknowledgement or last-contact state and mapping version. Method: trace the instruction through each enforcement layer, calculate confirmed and uncertain coverage and state a time-bounded lower and upper bound on enforcement. Uncertainty: offline controllers, stale group mapping, mobile-platform opacity, delayed queues, multiple linked tokens and acknowledgement semantics that do not prove physical effect. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: treat every unacknowledged component as UNKNOWN, make no further lifecycle change and escalate residual exposure to the incident authority.

ANSWER BOUNDARY

It must not infer where the credential is or whether it was used, expand the incident scope, suspend or revoke anything, or disclose event history by default. It must not direct incident tactics, determine culpability, suspend, revoke or restore access, control doors or emergency modes, disclose protected incident or movement records, or promise containment.

WHY THIS IS USEFUL

The administrator can give incident command a precise containment map instead of a misleading single suspended/not-suspended flag.

AC-062

QUESTION 2 OF 10

STAKEHOLDER QUESTION

What collateral access impact would each authorised suspension option have on unrelated duties and dependencies?

SENSORS AND TELEMETRY

Proposed evidence and location: current credential and entitlement graph, authorised incident scope options, role and duty evidence, route and controller dependencies, essential-access exceptions and dry-run change impact. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the affected identity, token or account and only the owner-defined candidate scopes, including direct and nested groups, shared routes, lift floors, schedules and offline controllers. Timing/aggregation: when the incident authority asks for options and whenever source or controller state changes; show aggregate impact first and authorised case detail second; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate dependency graph completeness, test scope simulation with synthetic nested cases, compare configuration hashes and verify current owner evidence and controller coverage; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a technical impact model cannot decide risk, necessity or proportionality; shared groups and unmapped local rules can hide both collateral loss and residual access.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the incident authority and Security own containment choice; managers and area owners own essential duties; PACS operations owns technical dependencies; HR or Student Records owns current affiliation. Explicit authorised candidate scopes and current duty/area evidence outrank analyst-created options, observed activity, broad role assumptions and convenience. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to the incident authority, Security, duty owner, area owner or PACS operations; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: scenario comparison and bounded recommendation. Decision interval: on request from the incident authority and before any authorised action. Minimum evidence: defined candidate scopes, current entitlement graph, essential-duty evidence, affected assets and controllers, exceptions and current configuration versions. Method: calculate retained and removed permissions, shared dependencies, residual paths and unrelated duty impact for each authorised option; recommend only the technically narrowest fully modelled option. Uncertainty: hidden local rules, incomplete duty evidence, offline controllers, stale mappings, concurrent roles and emergency exceptions. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return NO RECOMMENDATION where critical dependencies are unknown and leave the containment decision entirely with incident authority.

ANSWER BOUNDARY

It must not choose the risk response, decide whose duties matter, create a suspension scope, infer behaviour or execute containment. It must not direct incident tactics, determine culpability, suspend, revoke or restore access, control doors or emergency modes, disclose protected incident or movement records, or promise containment.

WHY THIS IS USEFUL

The administrator can explain collateral and residual technical effects before the authorised decision maker chooses an option.

Incidents, suspension, revocation and recovery

AC-061 to AC-070

AC-063

QUESTION 3 OF 10

STAKEHOLDER QUESTION

Has this urgent leaver or affiliation-end instruction propagated to every online and offline access-control component?

SENSORS AND TELEMETRY

Proposed evidence and location: authoritative end instruction, identity and affiliation status, credential and entitlement lifecycle state, distribution queues, controller acknowledgements, local cache versions and last-contact evidence. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the ended affiliation and every linked credential or account within the authorised instruction, all direct and inherited permissions and each enforcing component. Timing/aggregation: from authoritative instruction through each lifecycle and controller update, with incident-style status until reconciliation completes; no movement-event review is required; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate identity and token linkage, compare source, central and controller versions, verify queue sequence, clock offsets and configured offline behaviour using controlled test evidence; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: source end time may differ from access-effective time under policy, another affiliation may remain valid, and an offline controller can retain an older state.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: HR, Student Records or collaborator registry owns affiliation end; Security owns urgent access instruction; IAM and credential office own accounts/tokens; PACS operations owns enforcement propagation. The explicit current Security instruction grounded in authoritative affiliation evidence outranks directory lag, manager email alone, central state without acknowledgement and absence of events. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to the affiliation owner, Security, IAM, credential office or PACS operations; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: reconciliation and propagation estimate. Decision interval: immediately after instruction and until every in-scope component is confirmed or formally risk-managed. Minimum evidence: stable identity and instruction IDs, scope and effective time, current affiliations, linked tokens/accounts, complete entitlement paths and per-component state. Method: trace each lifecycle action and controller path, calculate confirmed, queued, offline and conflicting states and separate legitimate continuing affiliations. Uncertainty: identity duplication, concurrent roles, offline panels, stale group graph, queue failure, local manual changes and policy-defined grace periods. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: keep unresolved components UNKNOWN and escalate them; do not infer enforcement from central status or event silence.

ANSWER BOUNDARY

It must not decide affiliation, expand the leaver scope, revoke credentials, disclose movement records or claim complete enforcement without component evidence. It must not direct incident tactics, determine culpability, suspend, revoke or restore access, control doors or emergency modes, disclose protected incident or movement records, or promise containment.

WHY THIS IS USEFUL

The administrator can identify residual technical enforcement gaps instead of treating an urgent central update as universally complete.

AC-064

QUESTION 4 OF 10

STAKEHOLDER QUESTION

What authority, prior-state evidence and independent checks are required before considering restoration after an erroneous suspension?

SENSORS AND TELEMETRY

Proposed evidence and location: authoritative restoration workflow, incident finding and closure state, pre-suspension configuration snapshot, current identity and eligibility evidence, independent approvals, controller state and controlled post-change test plan. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the specific identity, credential and entitlements within the authorised restoration case, including changes that occurred legitimately during the suspension and every enforcing controller. Timing/aggregation: when restoration is proposed, immediately before authorised action and after propagation/test evidence; show a reconstruction plan, not an executable command; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: verify snapshot integrity and cut-off, compare current authority with prior state, test delta logic for intervening changes, validate independent approver roles and align all clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: blindly restoring the old snapshot can reintroduce expired or excessive access; incident records may remain disputed and offline controllers may hold a different state.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the incident authority owns the finding; HR, Student Records, sponsors and area owners own current eligibility; Security owns restoration approval; PACS operations owns implementation and test evidence. Fresh current authority and explicit independent restoration approval outrank the pre-incident state, user expectation, prior access events and a technical rollback capability. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: An approval or case record is stale when expired, superseded, outside its effective window, or missing a current owner/version; a connector state beyond the owner-approved freshness limit is UNKNOWN. Show conflicting values side by side and route them to the incident authority, current eligibility owner, Security or PACS operations; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: comparison and recommendation. Decision interval: after the incident decision, before authorised restoration and at post-change validation. Minimum evidence: verified incident outcome, current identity and eligibility, prior snapshot and provenance, intervening authorised changes, independent approvals, target controller list and test plan. Method: construct the current least-privilege state from fresh evidence, compare it with the prior snapshot, identify additions/removals caused by elapsed time and propose a bounded restoration delta. Uncertainty: incomplete prior snapshot, disputed incident outcome, expired prerequisites, role changes, offline controllers and unrecorded emergency changes. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return NO RESTORATION RECOMMENDATION until current authority and independent checks are complete; leave state unchanged.

ANSWER BOUNDARY

It must not determine that the suspension was erroneous, restore access, erase incident history or use the old state as automatic authority. It must not direct incident tactics, determine culpability, suspend, revoke or restore access, control doors or emergency modes, disclose protected incident or movement records, or promise containment.

WHY THIS IS USEFUL

The administrator can avoid both prolonged accidental denial and an unsafe full rollback to outdated permissions.

Incidents, suspension, revocation and recovery

AC-061 to AC-070

AC-065

QUESTION 5 OF 10

STAKEHOLDER QUESTION

Which revocation instructions remain queued for offline controllers, and which openings may still enforce stale access?

SENSORS AND TELEMETRY

Proposed evidence and location: authorised instruction queue, per-controller target set, last contact and heartbeat, local configuration version, queue age and retry state, with current opening and group mapping. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: all offline or intermittently connected controllers in the incident scope, their attached openings, relevant credentials or groups and the authoritative instruction cut-off. Timing/aggregation: at incident update intervals and on each reconnection or retry; report controller and opening counts, queue age and confidence, not person movements; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate target mapping and queue semantics with controlled tests, compare local and server versions after reconnect and verify controller and queue clocks; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: queued does not mean stored or enforceable, controller memory can be full, mappings may have changed and an opening's physical security cannot be inferred from stale access state alone.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Security or incident authority owns risk response; PACS operations owns queues and controller state; Estates and guarding operations own physical mitigation; network owners own connectivity restoration. The authoritative instruction plus controller-specific evidence outrank central queued status, retry count, missing events and assumptions that a closed door is secure. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to the incident authority, PACS operations, Network, Estates or guarding owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: observation, dependency lookup and estimate. Decision interval: throughout the incident until every controller is reconciled or an approved alternative control is in place. Minimum evidence: instruction ID and scope, complete controller/opening map, queue record, last contact, local version, retry result and current approved mitigation owner. Method: calculate known state-enforcement exposure by controller and opening, estimate uncertainty from last contact and queue capability, and separate technical from physical mitigation. Uncertainty: unknown local cache, changed mappings, queue rollover, network monitoring gaps, manual local changes and uninstrumented locks. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: treat the opening's enforcement as UNKNOWN, recommend escalation to the incident owner and make no remote action claim.

ANSWER BOUNDARY

It must not promise physical containment, direct guards, control doors, infer credential use or send the queued instruction. It must not direct incident tactics, determine culpability, suspend, revoke or restore access, control doors or emergency modes, disclose protected incident or movement records, or promise containment.

WHY THIS IS USEFUL

The administrator can identify where technical containment is uncertain so authorised owners can apply proportionate alternative controls.

AC-066

QUESTION 6 OF 10

STAKEHOLDER QUESTION

For this lockdown, controlled-unlock or emergency-mode request, who has authority and what contemporaneous evidence must be captured?

SENSORS AND TELEMETRY

Proposed evidence and location: incident-command authorisation record, approved emergency procedure reference, mode-command and acknowledgement logs, controller and opening state telemetry where fitted, fire/lift interface status and common-clock timeline. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: only the assets and systems explicitly named by incident command, including command issuer, executor, affected controllers/openings, integrations and rollback or recovery authority. Timing/aggregation: in real time from authorised command through acknowledgements and recovery, with immutable event-level evidence restricted to incident roles and aggregate status elsewhere; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: verify command source and role, procedure version, asset mapping, clock offsets and acknowledgement semantics; physical state checks occur only under competent incident procedure; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: system acknowledgements do not prove every physical door state; emergency commands can interact with fire, lift, power and local manual overrides, and telemetry may fail during the incident.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: incident command and approved emergency procedure own authority; Security executes authorised PACS actions; competent fire, lift, electrical and Estates owners own safety-critical dependencies. Current incident command and approved procedure outrank normal access policy, analytics, standing assumptions and any historical floor plan or event pattern. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to incident command and the competent system owners; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: authority lookup and contemporaneous observation. Decision interval: at request receipt, before execution by authorised staff, throughout the mode and during recovery handover. Minimum evidence: verified incident authority and role, procedure and scope, exact assets, command and acknowledgement IDs, clock quality, dependency status and recovery owner. Method: validate authority and scope, record the technical evidence chain and expose confirmed, unconfirmed and conflicting component states without recommending command strategy. Uncertainty: communications loss, clock failure, partial acknowledgement, manual overrides, safety-interface interaction and incomplete physical-state sensing. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: make no command through the service, return AUTHORITY UNKNOWN where verification fails and defer entirely to incident procedure.

ANSWER BOUNDARY

It must not initiate, approve or optimise emergency modes, advise incident tactics, override competent professionals or claim that telemetry certifies physical state. It must not direct incident tactics, determine culpability, suspend, revoke or restore access, control doors or emergency modes, disclose protected incident or movement records, or promise containment.

WHY THIS IS USEFUL

The administrator can support an auditable handover while keeping emergency authority and physical control outside the analytical service.

Incidents, suspension, revocation and recovery

AC-061 to AC-070

AC-067

QUESTION 7 OF 10

STAKEHOLDER QUESTION

Which PACS changes made through a suspected compromised administrator account require independent revalidation or rollback planning?

SENSORS AND TELEMETRY

Proposed evidence and location: privileged audit logs, approved change register, configuration snapshots and diffs, account and session metadata, controller acknowledgements, signed deployment evidence and service-account dependency records. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the suspected account and authorised incident window, every configuration object, identity, credential, schedule, integration and controller it changed or attempted to change. Timing/aggregation: from the incident-defined start through containment and investigation cut-off; present object-level diffs to authorised responders and aggregate impact outside the case; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: verify audit-log integrity and retention, align clocks, link commands to configuration diffs and acknowledgements, compare with approved change IDs and validate snapshot hashes; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: unapproved-looking activity is not proof of compromise; shared service accounts, automation and emergency changes can obscure attribution, and missing logs can understate impact.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Information Security and incident authority own compromise determination; Security and PACS service owners own access-control impact; change owners own approved work; auditors own evidence requirements. The incident scope and independent comparison with approved changes outrank behavioural anomaly scores, unusual timing, account reputation and assumptions from a username. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to Information Security, incident authority, PACS service owner, change owner or auditor; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: comparison and impact calculation. Decision interval: during incident scoping, after new evidence and before any authorised rollback decision. Minimum evidence: incident window and account, complete privileged logs, approved change register, pre/post snapshots, controller evidence, automation identities and log integrity status. Method: reconcile each observed change with an approved record, classify authorised, disputed, unexplained or unverifiable and calculate transitive dependencies and rollback prerequisites. Uncertainty: log gaps, clock skew, service-account indirection, out-of-band emergency work, partial controller acknowledgement and legitimate concurrent changes. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: preserve evidence, mark unexplained items for independent review and make no rollback or compromise conclusion.

ANSWER BOUNDARY

It must not declare the account compromised, expose secrets or session contents, attribute intent, disable the account or execute rollback. It must not direct incident tactics, determine culpability, suspend, revoke or restore access, control doors or emergency moves, disclose protected incident or movement records, or promise containment.

WHY THIS IS USEFUL

The administrator can produce an independent revalidation list focused on configuration evidence rather than unsupported behavioural inference.

AC-068

QUESTION 8 OF 10

STAKEHOLDER QUESTION

If this credential batch is defective or mis-encoded, which tokens and commissioned readers fall within the technical impact boundary?

SENSORS AND TELEMETRY

Proposed evidence and location: encoder batch and profile records, token serial ranges, personalisation station version, controlled read tests, credential technology metadata, reader model/firmware inventory and issue/return custody records. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the suspected batch, every token produced from it, issue status, affected credential technology and every commissioned reader type expected to process that encoding profile. Timing/aggregation: at defect report, during controlled sampling and before replacement planning; report batch and reader-class impact, with item-level detail restricted to authorised lifecycle staff; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: use approved reference and sample credentials, verify encoder calibration/self-test, inspect profile hash, test representative reader models and align issue, encode and test times; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a failed sample may reflect reader configuration, card damage or handling; a passed sample cannot prove every token, and reader conditions vary across installations.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the credential office owns batch and issue records; Security Systems owns reader estate; vendor or approved maintainer owns encoding and compatibility diagnosis; Security owns response scope. Controlled batch evidence and commissioned reader inventory outrank anecdotal denials, vendor generalisation, a single failed card and assumptions from serial adjacency alone. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to the credential office, Security Systems, approved maintainer, vendor or Security; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: diagnosis, sampling calculation and impact estimate. Decision interval: on defect report, after each controlled test stage and before an authorised remediation decision. Minimum evidence: batch ID and profile, complete serial range, issue status, reference test result, sampling plan, reader inventory and relevant firmware/configuration versions. Method: compare batch and reference results across representative reader classes, calculate supported lower and upper impact bounds and identify untested combinations. Uncertainty: small sample, heterogeneous reader tuning, damaged individual cards, profile drift, incomplete inventory and tokens already returned or destroyed. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: quarantine recommendations remain non-executable; return UNKNOWN beyond tested coverage and route broader action to Security and the credential office.

ANSWER BOUNDARY

It must not declare misuse, identify where a token was presented, disable a batch, replace credentials or adjust reader tuning. It must not direct incident tactics, determine culpability, suspend, revoke or restore access, control doors or emergency modes, disclose protected incident or movement records, or promise containment.

WHY THIS IS USEFUL

The administrator can bound the technical problem before launching an unnecessarily broad or too-narrow replacement programme.

Incidents, suspension, revocation and recovery

AC-061 to AC-070

AC-069

QUESTION 9 OF 10

STAKEHOLDER QUESTION

While ordinary access is suspended, is there a current owner-approved escorted route for essential work?

SENSORS AND TELEMETRY

Proposed evidence and location: authoritative essential-work approval, escort procedure and availability record, current route and closure data, temporary route group/schedule design and controller health; no live person tracking is used. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the approved task, named destination, escort authority, current route alternatives, required controlled openings and the exact temporary period proposed by the incident owner. Timing/aggregation: on request, before each approved work period and after route, escort or incident change; present route-access readiness, not location or escort tracking; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate route and door mappings against current owner records, test temporary group with non-person credentials, verify hard expiry and controller clocks and confirm escort process outside PACS; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: access permission cannot establish that an escort is present, the route is safe or the work remains essential; closures and incident boundaries can change rapidly.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: incident authority owns whether work may continue; task and area owners own essential purpose; Security owns escort and access procedure; Estates, Accessibility and safety owners own current route conditions. Current incident and route-owner approval outrank normal entitlement, past access, shortest-path calculation, event history and the May 2021 floor plan. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: An approval or case record is stale when expired, superseded, outside its effective window, or missing a current owner/version; a connector state beyond the owner-approved freshness limit is UNKNOWN. Show conflicting values side by side and route them to the incident, task, escort, route, area or safety owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, graph reachability and recommendation. Decision interval: when an authorised alternative is requested and before each operational period. Minimum evidence: current incident permission, essential-task approval, accountable escort procedure, current route/closure evidence, exact schedule, dependencies and controller status. Method: check whether an owner-approved route can be represented with the smallest temporary access set, flag non-technical escort and safety conditions and quantify unknown dependencies. Uncertainty: changing cordons, unverified escort availability, inaccessible route, offline controllers, shared doors and operating conditions outside access-control visibility. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return NO SAFE ACCESS RECOMMENDATION where route or authority is not current; keep ordinary suspension unchanged.

ANSWER BOUNDARY

It must not decide that work is essential or safe, assign an escort, track the route, create the temporary access or override incident boundaries. It must not direct incident tactics, determine culpability, suspend, revoke or restore access, control doors or emergency modes, disclose protected incident or movement records, or promise containment.

WHY THIS IS USEFUL

The administrator can show whether a bounded technical access option exists while leaving the operational decision with incident command.

AC-070

QUESTION 10 OF 10

STAKEHOLDER QUESTION

Before incident closure, which credentials, groups, schedules, controllers, integrations and evidence gaps still require validation?

SENSORS AND TELEMETRY

Proposed evidence and location: post-incident configuration and credential snapshots, exception register, controller and integration health, queue and event-gap status, clock quality, controlled tests, change trail and unresolved case register. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: every access-control object changed, suspended, restored, operated in emergency mode or left uncertain during the incident, plus its dependent assets and authoritative owner. Timing/aggregation: at recovery checkpoints and formal closure cut-off; present object-level status to incident roles and aggregate residual counts to governance; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: compare pre-incident approved baseline, incident-authorised changes and current state; verify hashes, clocks, queues, mappings and representative functional tests with recorded provenance; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: the pre-incident state may no longer be appropriate, some gaps cannot be reconstructed, and restored connectivity does not prove event completeness or physical function.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: incident command owns closure; Security owns access state; PACS, IAM, credential, integration and Estates owners own their components; Data Protection and Records Management own evidence handling. Current authorised recovery state and owner acceptance outrank automatic baseline rollback, green dashboards, absence of alarms and assumptions that all temporary changes expired. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to incident command and each accountable component or evidence owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: reconciliation, comparison and recommendation. Decision interval: at each recovery handover and immediately before formal incident closure. Minimum evidence: incident scope and timeline, approved pre/incident/post states, complete affected-object list, current component health, unresolved gaps, tests, owner acceptance and evidence permissions. Method: build a closure matrix by object and owner, classify validated, intentionally changed, outstanding, irrecoverable gap or conflict and identify required follow-up without closing it. Uncertainty: missing baseline, unlogged emergency action, clock or event gaps, offline devices, unresolved identity decisions and safety evidence outside PACS. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: keep closure status OPEN or CONDITIONAL for critical unknowns, preserve evidence and defer acceptance to incident command and competent owners.

ANSWER BOUNDARY

It must not close the incident, erase temporary evidence, certify recovery, restore a baseline or disclose incident data beyond approved roles. It must not direct incident tactics, determine culpability, suspend, revoke or restore access, control doors or emergency modes, disclose protected incident or movement records, or promise containment.

WHY THIS IS USEFUL

The administrator can give incident command a complete access-control recovery checklist and prevent hidden temporary states from surviving closure.

Privacy, retention, governance and service reporting

AC-071 to AC-080

AC-071

QUESTION 1 OF 10

STAKEHOLDER QUESTION

For this stated purpose, who may view person-level access events, which fields and interval are permitted, and who approved access?

SENSORS AND TELEMETRY

Proposed evidence and location: data-access request workflow, purpose and recipient approval, data-catalog metadata, query field/time filters, role-based access checks and immutable audit logging before any person-level event retrieval. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the specific approved purpose, named case and recipients, minimum credential or event fields, exact time window and source systems; no data is retrieved until permission gates pass. Timing/aggregation: at request intake, immediately before query or export and at access expiry; report permission status and requested scope, not the protected events themselves; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: test role and purpose enforcement with synthetic records, verify query filters and audit-log completeness, align approval, query and retrieval clocks and record export provenance; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: technical access does not establish a lawful purpose, approvals may cover only part of the requested fields or period and small datasets can reveal movement even after names are removed.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the requesting authority owns purpose; Data Protection and Legal own lawful basis and disclosure conditions; Security owns operational sensitivity; system owners own access enforcement; Records Management owns retention. Current purpose-specific written authority and minimum-necessary decision outrank job seniority, system capability, prior access, urgency alone and a general investigation label. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to Data Protection, Legal, Security, the requesting authority or source-system owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: permission lookup and scope validation. Decision interval: at request, before every retrieval or disclosure and when scope changes. Minimum evidence: verified requester and role, documented purpose and lawful authority, named recipients, exact fields and period, minimisation decision, security classification and retention/transfer conditions. Method: validate each requested element against approval, reduce to minimum fields and interval, identify third-party or security-sensitive data and record the decision chain. Uncertainty: ambiguous purpose, overbroad recipient group, re-identification risk, mixed data ownership, legal hold and source records outside the approved window. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: retrieve or disclose nothing through the analytical service, return NOT AUTHORISED or PARTIALLY AUTHORISED and route unresolved scope to the named owner.

ANSWER BOUNDARY

It must not grant log access, reveal events while checking permission, infer movement or intent, expand purpose or treat technical administrator rights as disclosure authority. It must not determine lawful basis or exemptions, grant disclosure access, share or delete records, infer movement, attendance or productivity, handle secrets, certify governance or control access systems.

WHY THIS IS USEFUL

The administrator can decide whether a request is ready for an authorised data handler and exactly what minimum dataset may be considered.

AC-072

QUESTION 2 OF 10

STAKEHOLDER QUESTION

Which credential, administrative and door-event records are due for deletion, continued retention or legal hold now?

SENSORS AND TELEMETRY

Proposed evidence and location: data inventory and classification, owner-approved retention schedules, system retention configuration, deletion-job status, backup and archive inventory, legal-hold register and immutable action logs. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: each access-control data class and store in scope, including primary systems, replicas, exports, audit logs, backups and vendor platforms, with owner and jurisdiction. Timing/aggregation: at the policy-defined review interval, before scheduled deletion and immediately on an authorised hold or policy change; report counts and ages by data class; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate inventory against actual configured stores, test retention rules on synthetic data, verify deletion and hold job evidence, align clocks and preserve transformation and transfer provenance; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a configured deletion period does not prove all replicas are deleted, backups may follow separate rules and legal or incident holds can override routine schedules.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Records Management owns schedules; Data Protection and Legal own lawful requirements and holds; Security owns security-sensitive evidence; each system and vendor owner owns implementation. The current approved retention rule and explicit legal hold outrank default vendor settings, storage age, local practice, system capacity and assumptions from a missing record. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to Records Management, Data Protection, Legal, Security or the relevant system owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, comparison and calculation. Decision interval: at retention review, before deletion and on every hold or policy change. Minimum evidence: complete data-store inventory, data class and owner, current schedule and effective date, legal/incident holds, record age, backup path and deletion capability evidence. Method: map each store and data class to the controlling rule, calculate due actions, identify conflicts and distinguish configured, queued, evidenced complete and unverifiable deletion. Uncertainty: unknown replicas, vendor opacity, clock and time-zone differences, export copies, backup immutability, overlapping holds and incomplete inventory. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: execute no deletion or hold; mark the action UNKNOWN or CONFLICTED and route it to the authoritative records and system owners.

ANSWER BOUNDARY

It must not delete, retain or place records on hold, interpret law, conceal an evidence gap or disclose protected records during inventory. It must not determine lawful basis or exemptions, grant disclosure access, share or delete records, infer movement, attendance or productivity, handle secrets, certify governance or control access systems.

WHY THIS IS USEFUL

The administrator can prepare a store-by-store action list that avoids both premature deletion and indefinite default retention.

Privacy, retention, governance and service reporting

AC-071 to AC-080

AC-073

QUESTION 3 OF 10

STAKEHOLDER QUESTION

Which access-control records fall within this data-rights request, and how must third-party and security-sensitive content be protected?

SENSORS AND TELEMETRY

Proposed evidence and location: authorised rights-request case, data catalog and search index, identity-linkage outcome from the approved process, record-source inventory, third-party flags, security classification, redaction rules and export provenance. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the verified request scope and date range across credential, administrative and access-event systems, including copies and exports that the authoritative process identifies. Timing/aggregation: at scoped search, review and authorised export; return record counts and source coverage first, with content only to approved rights-response roles; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate search keys with controlled cases, verify source coverage and cut-off, test redaction on synthetic records, align clocks and record every query, exclusion and transformation; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: name or credential matching can miss aliases or include another person, records may contain third-party data and security exemptions require authoritative legal assessment.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the Data Protection rights team owns scope and identity verification; Legal owns exemptions; Security owns operational sensitivity; system owners own source retrieval; Records Management owns completeness expectations. The approved rights-request instruction and source-owner records outrank requester-supplied search guesses, administrator judgement, broad name matching and convenience. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to the Data Protection rights team, Legal, Security, Records Management or source-system owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, coverage calculation and redaction recommendation. Decision interval: at search design, after coverage review and before authorised response release. Minimum evidence: verified request and identity matching authority, exact scope and cut-off, complete source inventory, retention state, third-party/security classifications and approved redaction rules. Method: search each authorised source using approved identifiers, calculate coverage and gap statements, flag third-party or sensitive fields and prepare a provenance-preserving candidate set. Uncertainty: alias and duplicate identity, expired records, inaccessible backups, source gaps, shared credentials, clock uncertainty and exemptions not machine-determinable. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: release nothing through the service, present gaps and candidate records only to authorised response staff and route exemption decisions to Legal/Data Protection.

ANSWER BOUNDARY

It must not verify the requester, decide an exemption, reveal third-party data, infer movement or send the response. It must not determine lawful basis or exemptions, grant disclosure access, share or delete records, infer movement, attendance or productivity, handle secrets, certify governance or control access systems.

WHY THIS IS USEFUL

The administrator can support a complete, minimised search while keeping legal judgement and disclosure with the authorised rights team.

AC-074

QUESTION 4 OF 10

STAKEHOLDER QUESTION

Which service measures can be reported with valid denominators and uncertainty without exposing identities or movement patterns?

SENSORS AND TELEMETRY

Proposed evidence and location: case and task timestamps, queue states, credential inventory counts, controller and integration health, change volumes, fault categories and approved service-target definitions; person-level door events are excluded. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the defined reporting period and service population for requests, credentials, faults, changes and reviews, including paused time and known system downtime. Timing/aggregation: daily operational aggregation and monthly or owner-defined reporting; suppress small cells and retain case detail only for authorised service management; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate event-to-metric mapping with sample cases, reconcile opening/closing populations, test pause and denominator rules, align clocks and tag missing or delayed source intervals; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: average time can hide queues and exclusions, counts without eligible population are misleading, and system downtime or policy pauses can distort comparisons.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the access-control service owner owns operational metrics; governance owns targets; Data Protection owns aggregation and disclosure rules; source owners own case and health data. The current metric definition, eligible denominator and complete reporting window outrank easy-to-count events, raw transaction volume, last year's formula and informal benchmarks. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to the service owner, governance, Data Protection or relevant source owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: calculation and comparison. Decision interval: at operational review and formal reporting cut-offs. Minimum evidence: approved metric definitions, numerator and denominator populations, period and time zone, pause rules, target, source coverage, known outages and suppression threshold. Method: calculate distributions and percentiles where appropriate, show denominators and missing coverage, separate demand from completion and compare only like definitions and periods. Uncertainty: small populations, changing definitions, incomplete case closure, reopened work, paused cases, clock gaps and source downtime. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: suppress unsafe cells, mark incomplete measures PARTIAL and omit comparisons whose definitions or coverage are not equivalent.

ANSWER BOUNDARY

It must not rank individuals, infer attendance or productivity, publish movement counts, expose small identifiable groups or certify service quality from incomplete metrics. It must not determine lawful basis or exemptions, grant disclosure access, share or delete records, infer movement, attendance or productivity, handle secrets, certify governance or control access systems.

WHY THIS IS USEFUL

The administrator can report defensible workload, timeliness and reliability measures that support staffing and process decisions without surveillance.

Privacy, retention, governance and service reporting

AC-071 to AC-080

AC-075

QUESTION 5 OF 10

STAKEHOLDER QUESTION

What lawful authority, approvals, minimisation and secure-transfer controls are required before external disclosure of access-control records?

SENSORS AND TELEMETRY

Proposed evidence and location: external-disclosure workflow, recipient and purpose approval, data classification, approved field/time filter, redaction status, secure-transfer channel health, delivery acknowledgement and immutable disclosure log. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the specific recipient, purpose, legal or contractual authority, minimum record set, time window, transfer route, retention condition and any onward-sharing restriction. Timing/aggregation: at request, before package creation, immediately before transfer and on receipt acknowledgement; expose only readiness and package metadata until authorised release; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: test recipient and role validation, redaction and field filters with synthetic data, verify encryption and channel configuration without exposing keys, align clocks and record file hash and custody; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a secure channel does not establish lawful sharing, recipient identity can change, redaction can fail through linked fields and onward copies may fall outside technical control.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the requesting authority owns purpose; Data Protection and Legal own lawful basis and conditions; Information Security owns transfer controls; Security owns operational sensitivity; Records Management owns retention. Current written disclosure authority and minimum-necessary decision outrank urgency, recipient reputation, prior sharing, technical encryption and administrator access. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to Data Protection, Legal, Information Security, Security or Records Management; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: permission lookup and validation. Decision interval: at each disclosure stage and whenever recipient, purpose, fields or channel change. Minimum evidence: verified recipient and authority, approved purpose, exact data scope and time window, classification, redaction, transfer method, retention/return conditions and accountable approvers. Method: validate every gate, minimise and hash the package, record chain of custody and identify any condition the platform cannot enforce. Uncertainty: recipient ambiguity, re-identification, mixed ownership, channel failure, third-party data, onward sharing and inconsistent retention rules. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: create or transfer no package through the service; return NOT AUTHORISED or NOT READY and route the exact failed gate.

ANSWER BOUNDARY

It must not decide lawful basis, reveal records while assessing the request, handle encryption keys, send the package or assume delivery acknowledgement ends responsibility. It must not determine lawful basis or exemptions, grant disclosure access, share or delete records, infer movement, attendance or productivity, handle secrets, certify governance or control access systems.

WHY THIS IS USEFUL

The administrator can prepare an auditable disclosure gate that separates lawful authority, data minimisation and secure transfer.

AC-076

QUESTION 6 OF 10

STAKEHOLDER QUESTION

Which privacy, accessibility, security and governance gates must be passed before adopting mobile or other identity-linked access technology?

SENSORS AND TELEMETRY

Proposed evidence and location: approved architecture and data-flow inventory, DPIA or equivalent status, security assessment, accessibility evaluation, compatibility and resilience tests, vendor/processor records and commissioning evidence. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the proposed technology, every personal-data flow and identifier, device and reader classes, fallback path, supplier service, integration, data location and lifecycle stage. Timing/aggregation: at design gates, pilot, material change and before production acceptance; report control status and test coverage, not user behaviour or device contents; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate data flows against observed test traffic metadata under approval, test representative devices/readers and failure modes, verify accessibility methods, clocks, logs and supplier configuration provenance; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: a pilot cannot cover all users or conditions, vendor claims may be incomplete, device updates can invalidate evidence and identity-linked telemetry can create new tracking risk.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Data Protection owns privacy assessment; Information Security owns security acceptance; Accessibility and Equality owners own inclusive design; Security and PACS owners own access operation; Procurement and Legal own supplier terms. Current signed governance decisions and independent test evidence outrank vendor marketing, a successful demonstration, technical convenience and assumed user consent. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to Data Protection, Information Security, Accessibility, Security, Procurement, Legal or the PACS owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: comparison and recommendation. Decision interval: at each formal design, pilot and production acceptance gate. Minimum evidence: documented purpose and necessity, data-flow map, lawful basis, minimisation, security and accessibility evidence, compatibility/resilience tests, fallback, supplier terms and accountable owners. Method: compare the proposal with approved requirements, identify failed or untested controls and recommend proceed-to-next-gate, hold or redesign without authorising adoption. Uncertainty: unrepresented users, supplier opacity, device drift, inaccessible fallback, offline operation, data-location change and future feature expansion. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return NO GO RECOMMENDATION for critical unknowns, preserve a non-digital fallback and require authoritative owner decisions.

ANSWER BOUNDARY

It must not approve procurement or deployment, access device contents, handle secrets, force identity-linked technology or certify privacy, accessibility or security. It must not determine lawful basis or exemptions, grant disclosure access, share or delete records, infer movement, attendance or productivity, handle secrets, certify governance or control access systems.

WHY THIS IS USEFUL

The administrator can prevent a technically attractive credential method from entering service without complete governance and fallback evidence.

Privacy, retention, governance and service reporting

AC-071 to AC-080

AC-077

QUESTION 7 OF 10

STAKEHOLDER QUESTION

How complete, current, mapped and healthy is the evidence estate across openings, controllers, credentials, identity feeds and administrative logs?

SENSORS AND TELEMETRY

Proposed evidence and location: asset and data-source inventory, mapping completeness, commissioning and last-test status, heartbeat and uptime, clock-offset health, schema/version state, provenance coverage, permission controls and known gap register. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: every evidence source used by the catalogue, including physical openings and sensors where fitted, controllers, credential platforms, authoritative feeds, workflows, logs, integrations and exports. Timing/aggregation: live health at source-specific heartbeat, daily quality aggregation and formal monthly or owner-defined assurance snapshot; report by source and evidence dimension; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate inventory against physical and platform discovery under approval, sample mappings, run known test events, verify calibration or functional-test dates, clock offsets, gap detection and lineage; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: inventory itself can be incomplete, green health does not prove semantic correctness, and sources have different coverage, sampling, retention and maintenance regimes.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: each source-system or asset owner owns current truth and health; Security owns access-control evidence requirements; Data Protection and Information Security own permission and protection; governance owns assurance thresholds. Owner-verified inventory, commissioning, health and provenance evidence outrank self-reported integration claims, dashboard availability and the May 2021 floor plan. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A live status is stale after two expected heartbeats or the asset owner's approved alarm threshold; buffered, duplicated or late events remain explicitly tagged. Show conflicting values side by side and route them to the relevant source owner, Security, Data Protection, Information Security or governance owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: observation and calculation. Decision interval: continuously for health and at each formal evidence-assurance cut-off. Minimum evidence: complete source inventory, owner, purpose and permission, coverage, expected cadence, commissioning/test status, clock quality, uptime/gap metrics, version and lineage. Method: score each evidence dimension separately, calculate known coverage and gap intervals and avoid a single composite score that hides critical unknowns. Uncertainty: undiscovered assets, unmonitored gaps, different sampling, stale commissioning, vendor opacity, incomplete permissions and semantic drift despite uptime. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: mark the affected analytical answers PARTIAL or UNKNOWN and create an owner-specific evidence backlog rather than substituting another weak source.

ANSWER BOUNDARY

It must not claim deployment from documentation, certify evidence fitness universally, use data beyond approved purpose or treat the historical floor plan as current truth. It must not determine lawful basis or exemptions, grant disclosure access, share or delete records, infer movement, attendance or productivity, handle secrets, certify governance or control access systems.

WHY THIS IS USEFUL

The administrator can prioritise commissioning, mapping and integration work where it most affects dependable access decisions.

AC-078

QUESTION 8 OF 10

STAKEHOLDER QUESTION

Are access requests, credential tasks and fault cases meeting agreed service targets, and where is avoidable delay accumulating?

SENSORS AND TELEMETRY

Proposed evidence and location: case creation, queue, assignment, pause, escalation and closure timestamps, task type, dependency wait states, integration downtime and owner-approved service target definitions. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: all eligible access request, credential and technical fault cases in the reporting period, with defined exclusions, reopened cases and known periods of system outage. Timing/aggregation: daily queue observation and monthly or agreed performance reporting; aggregate by work type, stage and accountable dependency, with small-cell suppression; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate workflow-state transitions with sampled cases, reconcile open/closed populations, test pause and target logic, align clocks and tag missing intervals and backfilled updates; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: elapsed time does not equal controllable service delay, cases differ in complexity and paused time can be legitimate or inconsistently recorded.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the service owner owns process measures; governance owns targets; source-system owners own case events; Data Protection owns aggregation; dependency owners own external wait reasons. Current agreed target definitions and complete eligible populations outrank raw averages, ticket counts, individual rankings and comparisons using different workflow rules. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: A policy, matrix or retention rule without a current owner, effective date and version is stale; pending drafts and superseded instructions cannot authorise action. Show conflicting values side by side and route them to the service owner, governance, Data Protection or dependency owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: calculation, comparison and diagnosis. Decision interval: at operational review and formal reporting cut-off. Minimum evidence: approved case population and cut-off, target and pause rules, complete state history, work type, dependency waits, reopened status, source coverage and outage intervals. Method: calculate median and percentile elapsed and active handling intervals, stage queues and ageing, compare like periods and locate process stages with persistent evidence-supported delay. Uncertainty: inconsistent pauses, complex cases, missing transitions, reopened work, changing target definitions, small groups and system downtime. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: publish no person-level ranking, mark incomplete stages PARTIAL and avoid causal claims where the data only locates accumulation.

ANSWER BOUNDARY

It must not infer staff productivity or intent, expose requester or administrator identities, promise resolution times or use access events as workload evidence. It must not determine lawful basis or exemptions, grant disclosure access, share or delete records, infer movement, attendance or productivity, handle secrets, certify governance or control access systems.

WHY THIS IS USEFUL

The administrator can direct process improvement or staffing attention to queues and dependency stages supported by defensible measures.

Privacy, retention, governance and service reporting

AC-071 to AC-080

AC-079

QUESTION 9 OF 10

STAKEHOLDER QUESTION

How can this denial be explained accurately without revealing protected security configuration or another person's data?

SENSORS AND TELEMETRY

Proposed evidence and location: the authorised case's deny reason, entitlement and schedule status, reader/controller health, current mode, approved explanation taxonomy and redaction rules; no unrelated credential transactions are retrieved. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the single reported transaction or failure window, affected credential and reader, current access case and only the configuration facts permitted for requester communication. Timing/aggregation: on case review after technical evidence is current; generate a minimal explanation and expiry time for that explanation if the state may change; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: validate deny-code mapping, reader/door identity and clock offset, compare configuration version and test redaction templates against synthetic cases; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: deny codes can be ambiguous or vendor-specific, technical state may have changed since the event and revealing group, zone or mode detail can create security risk.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: Security owns disclosure boundaries and explanation taxonomy; PACS operations owns technical interpretation; the sponsor or area owner owns eligibility; Data Protection owns personal-data minimisation. Current authorised explanation rules and verified case evidence outrank a raw vendor code, another person's outcome, speculative cause and requester pressure for configuration detail. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to Security, PACS operations, the eligibility owner or Data Protection; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: lookup, diagnosis and redaction. Decision interval: when a requester-facing explanation is needed and after material state change. Minimum evidence: verified case, exact reader/time with clock quality, current entitlement and schedule, deny-code interpretation, health state and approved disclosure template. Method: select the least revealing accurate explanation category, state whether the cause is confirmed or unknown and direct the requester to the accountable next step. Uncertainty: stale event context, ambiguous deny code, hidden local mode, unresolved eligibility, shared credentials and redaction that could still reveal sensitive layout or policy. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return a neutral UNKNOWN/UNDER REVIEW message and route internally; disclose no configuration or other-person data.

ANSWER BOUNDARY

It must not reveal protected door groups, schedules, other users, movement records or incident states, infer intent, promise entry or change access to resolve the complaint. It must not determine lawful basis or exemptions, grant disclosure access, share or delete records, infer movement, attendance or productivity, handle secrets, certify governance or control access systems.

WHY THIS IS USEFUL

The administrator can communicate a useful next step while protecting security-sensitive configuration and third-party information.

AC-080

QUESTION 10 OF 10

STAKEHOLDER QUESTION

What approvals, design records, tests, ownership, privacy controls and handover evidence are required before onboarding this opening or integration?

SENSORS AND TELEMETRY

Proposed evidence and location: asset and design records, approved data and control-flow diagram, PACS object and controller mapping, commissioning tests, reader/lock/contact/REX and interface results where fitted, clock/health monitoring, privacy and security approvals and handover register. Commissioning gate: use only after source and asset mapping, purpose permission, functional commissioning and current health are independently verified. Coverage: the complete new opening or integration, every physical and digital dependency, data field, controller, network, power, fire/lift interface, permission group, owner, maintenance path and fallback. Timing/aggregation: at design, pre-commissioning, commissioning, acceptance and material change; retain asset-level test evidence and publish only readiness and unresolved-gate status; retain native event time and receive time, and expose person-level detail only inside an authorised case, otherwise case, asset or interval aggregates. Quality, calibration and provenance: physically verify asset IDs, use calibrated technician instruments where required, approved test credentials and witnessed state tests, compare design/as-built/configuration, verify clocks, gaps, firmware and provenance; record the approved clock source and tolerance, uptime, gaps, duplicate or late events, drift and source-to-answer lineage. Limits: successful component tests do not certify the whole system or current room use; hidden dependencies, incomplete ownership and later works can invalidate handover evidence.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence: the project and Estates owners own design/as-built and assets; competent fire, lift, electrical and security professionals own their interfaces; Security owns access policy; Data Protection and Information Security own data/control governance; PACS operations owns service acceptance. Current signed approvals, as-built records, competent tests and named operational ownership outrank design intent, vendor demonstration, a central green status and the May 2021 floor plan. Evidence discipline: preserve effective time, observed or approved time and retrieval time separately, together with purpose, authorised role, owner, source system, record or version ID and transformation lineage. Person-level detail is limited to the approved case. Staleness and conflict: Commissioning or maintenance evidence is stale after the owner-defined retest interval, after relevant works or configuration change, or when the tested asset mapping cannot be reproduced. Show conflicting values side by side and route them to the project, Estates, competent interface, Security, Data Protection, Information Security or PACS service owner; Do not merge, overwrite or resolve them silently.

ANALYSIS REQUIRED

Operation: checklist comparison and recommendation. Decision interval: at each acceptance gate and after any material design or configuration change. Minimum evidence: unique asset IDs, approved purpose and ownership, current design/as-built, access and integration mapping, test plan/results, clock/health evidence, privacy/security decisions, maintenance and fallback. Method: compare every mandatory gate with current evidence, identify missing or conflicting ownership and test coverage and recommend accept-to-next-gate, hold or rework without commissioning the asset. Uncertainty: partial physical tests, vendor opacity, unverified interfaces, undocumented local configuration, missing maintenance owner, accessibility gaps and post-test works. Output one evidence state: VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. Safe failure: return NOT READY for any critical missing gate, leave the opening or integration outside analytical service scope and route to the accountable owner.

ANSWER BOUNDARY

It must not commission, connect or control the opening or integration, certify safety/security/privacy, decide room use, handle secrets or approve production service. It must not determine lawful basis or exemptions, grant disclosure access, share or delete records, infer movement, attendance or productivity, handle secrets, certify governance or control access systems.

WHY THIS IS USEFUL

The administrator can enforce a complete evidence and ownership gate before a new asset becomes an unsustainable or poorly governed part of the access estate.

Operational summary

What the catalogue helps an access-control administrator do

1

Identity and role-based access requests

Validate identity, role, sponsor and prerequisite evidence; prepare the minimum defensible request package.

AC-001 to AC-010

2

Credential issue, replacement and lifecycle

Issue, replace and reconcile credentials through controlled custody, test and lifecycle evidence.

AC-011 to AC-020

3

Doors, zones, permission groups and dependencies

Trace opening, zone, group, lift, schedule and safety-critical dependencies before configuration work.

AC-021 to AC-030

4

Schedules, temporary access and visitors

Design bounded recurring and temporary windows for duties, visitors, contractors, closures and events.

AC-031 to AC-040

5

Reader, controller and integration faults

Locate reader, controller, network, power, clock and integration fault candidates with controlled tests.

AC-041 to AC-050

6

Reviews, audit trails and least privilege

Run complete entitlement reviews, expose unsupported privilege and preserve reproducible change chains.

AC-051 to AC-060

7

Incidents, suspension, revocation and recovery

Show containment, propagation and recovery uncertainty without taking command or changing access.

AC-061 to AC-070

8

Privacy, retention, governance and service reporting

Protect person-level records, enforce retention and permission gates, and report aggregate service evidence.

AC-071 to AC-080

SAFE DECISION PATTERN

1

Identify authority

2

Verify current evidence

3

State coverage and uncertainty

4

Perform one bounded operation

5

Return a non-executing next step

Evidence and permissions rules

Rules that apply to all 80 questions

1

Proposed until verified

No sensor, integration or analytical capability is treated as deployed until mapping, commissioning, purpose permission and current health are independently established.

2

Authority before telemetry

Identity, affiliation, sponsorship, eligibility, safety, incident and data-protection decisions come from current accountable owners, not sensor observations.

3

Historical plan only

The May 2021 floor plan may help reconcile legacy identifiers. It cannot establish current room use, access, services, hazards, accessibility, routes or closures.

4

Three times

Keep effective, observed or approved, and retrieved times distinct. Never replace source time with retrieval time or display order.

5

Measured coverage

State asset, location, interval, sampling or aggregation, uptime, gaps and unobserved conditions for every sensor or log source.

6

Quality and provenance

Verify calibration or controlled functional tests, clock alignment, version, drift, duplicates, latency, transformation and source-to-answer lineage.

7

No passage inference

A credential transaction is a system decision. It is not proof of identity, passage, attendance, occupancy, behaviour, safety or intent.

8

Minimum necessary data

Use person-level records only for a verified purpose and authorised case. Prefer configuration, case status and safe aggregates.

9

Conflict stays visible

Present current conflicting sources side by side with owners and times. Do not silently merge, rank or overwrite authoritative records.

10

Uncertainty is an output

Use VERIFIED, PARTIALLY VERIFIED, CONFLICTED or UNKNOWN. State the reason, affected coverage and evidence that would change the state.

11

Safe failure

When minimum evidence, permission, mapping or health is absent, keep access unchanged and route the precise gap to its accountable owner.

12

Decision support only

The service does not grant, modify, suspend, revoke, restore or expire access; handle secrets; disclose protected records; certify safety or security; or control systems.